



History of the concept of nuclear shape

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Abstract This article provides a comprehensive review of the evolution of the nuclear shape concept, a cornerstone in nuclear physics. Tracing its historical development from the early 20th century, we highlight key milestones and paradigm shifts that have shaped our understanding. The review explores the transition from the initial spherical model to the introduction of nuclear deformation, emphasizing the contributions of the liquid drop model and the unified model. The pivotal role of nuclear shapes in elucidating various nuclear phenomena and their profound impact on both theoretical and experimental nuclear physics are discussed in depth. The article underscores the relevance of nuclear shape in contemporary physics, particularly in light of ground-breaking findings from ultra-relativistic heavy ion collisions. These recent results illustrate the enduring significance of nuclear shape in advancing our comprehension of nuclear structure and reactions.

“If I had a great calculating machine, I would perhaps apply it to the Schrödinger equation of each metal and obtain its cohesive energy, its lattice constant, etc. It is not clear, however, that I would gain a great deal by this. Presumably, all the results would agree with the experimental values and not much would be learned from the calculation. What would be preferable, instead, would be a vivid picture of the behavior of the wave function”

E. P. Wigner quoted by E. Guth at the Conference on Bases for Nuclear Spin-Parity Assignments,

Gatlinburg, Tennessee (1965) [1]

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1 Introduction

The coherent understanding of the internal structure of the atomic nucleus, the excitation (and de-excitation) regimes it undergoes—meaning its ability to emit and absorb radiation—and its reaction mechanisms from the accelerators to the stars, is often regarded as one of the most ambitious tasks humanity has set for itself. This is linked to its unique nature, often described as distinctive (“the nucleus is a unique laboratory for...”): a quantum object composed of several nucleons interacting via a complex residue of the strong interaction, strongly correlated and giving rise to a rich phenomenology. To tackle this formidable endeavor, numerous concepts have been forged throughout the history of nuclear physics. Some were inspired by other domains of older physical studies, such as shell structure or pairing, while others have been custom-made, often challenging the paradigmatic views of their time.

The concept of nuclear shape or deformation is a prime example of the latter. It is a perfect illustration of how the rich phenomenology associated to the atomic nucleus has genuinely compelled physicists to introduce a form of creativity into the quantum realm. No one truly expected to find themselves in such a position because, fundamentally, those who embarked on the journey into the infinitesimally small were propelled by an unspoken yet distinctly Platonic desire to trace the essential causes of the vibrant richness of the perceptible world back to an underlying order, an elemental realm governed by symmetries (an approach that can be qualified as reductionist).

So, on this trajectory towards the infinitesimally small in search of a more regular and perfect underlying order, there stood the nucleus... Naturally, when encountered, when it was understood to have a finite dimension, the question of the shape of this spatial extension did not arise; it defaulted to the perfect shape, the sphere—the absence of shape, in fact! We will see that until an advanced stage in the evo-

lution of nuclear physics, around the late 1930s, the notion that the nucleus was spherical remained universally prevalent (indeed, echoes of this notion are still found in Feaser's monograph [2] as late as 1952).

However, the idea that the nucleus could possess a shape eventually became a necessity as measurable nuclear quantities multiplied, the precision of these measurements improved, and new modes of nuclear excitation were discovered. The concept of nuclear shape immediately allowed us to encompass a significant portion of what is now referred to as the “emergent” properties of nuclear systems. Emergence can be seen as a universal characteristic of complex systems [3]. Paradoxically, however, the reality and tangibility of the nuclear shape are questionable. It is important to emphasize at this point that there is no operational definition of any parameter that would serve as a measure of nuclear shape. Defining a shape for the nucleus involves attaching an intrinsic reference frame, yet experiments are conducted in the laboratory frame.

This dual characteristic of the concept of nuclear shape, towards which a vast and diverse phenomenology points without the nuclear shape itself ever being directly accessible, was elegantly illustrated by Fig. 1 in the first review on the subject, credited to Temmer [4]. I propose here a reinterpretation of this figure (see Fig. 15 in Sect. 4), expanding it to include this new branch opening up in the realm of relativistic heavy-ion collisions [5–7], which is the subject of this series of articles, including this one.

It is hoped that by the end of this article, the reader will be convinced that the concept of nuclear shape, along with all associated developments and refinements, has throughout the history of nuclear physics facilitated the interpretation (some would dare say “understanding”) of numerous nuclear phenomena and correlated multiple observables of profoundly different origins. Yet, even for us physicists who have halted our exploration at the nuclear scale in the quest for the infinitesimally small, the considered correct way to account for all nuclear phenomena is the so-called “microscopic” approach. The ultimate goal is to describe these phenomena starting from the individual degrees of freedom of N interacting bodies (which are, however, no longer truly protons and neutrons, a kind of local reductionism, or effective separation of degrees of freedom).

The concept of shape, seemingly the only one borrowed from the macroscopic world in the entire history of nuclear physics¹ (as everyone agrees that everyday objects have a shape), evidently clashes with this approach. Indeed, the notion that the nucleus possesses a shape, and thus a sur-

face capable of oscillation and rotation, opened the way for a certain “macroscopic” description of its excitations, portraying them as potentially “collective,” as if the nucleus behaved as a homogeneous whole. It is often considered, following the now-historic success of the so-called “unified” model by Bohr and Mottelson, that microscopic and macroscopic views of the nucleus have long been reconciled and are no longer a real issue for contemporary nuclear structure. However, in hindsight, it is evident that this reconciliation was achieved only by shifting the problem to understanding the microscopic origins of collective phenomena, the microscopic reasons for specific shape transition or coexistence phenomena, for instance, or the microscopic origin of low-energy satellites of a giant resonance, while that last is supposed to exhaust the strength of an operator applied to the total wave function of the nucleus, etc. From this perspective, it seems interesting to address the peculiar relationship between a specific “microscopic” model, namely the shell model, and the concept of nuclear shape—relationships much deeper than one might think because the latter emerged by drawing on the former (which, however and paradoxically enough, directly describes the nucleus in the laboratory frame).

Overall, we will see that the introduction of the concept of nuclear shape is one of the most remarkable events in the history of nuclear structure. This concept has allowed us to both account for a vast phenomenology and correlate numerous observables from measurements of entirely different natures. These two qualities are the hallmark of very good scientific concepts—effective concepts. In fact, the Nobel Committee of the time recognized this, awarding the Nobel Prize in Physics in 1975 to the main contributors, creators, and developers of the nuclear shape concept: L. J. Rainwater, A. N. Bohr, and B. R. Mottelson.

Nevertheless, such effectiveness always comes with a cost, and the introduction of the concept of nuclear shape has brought about a series of paradoxes and theoretical and conceptual difficulties. Its incredible utility in analyzing data from relativistic heavy-ion collisions, and the information on nuclear shapes that we can expect in return, demonstrate the incredible vitality of this concept. However, it remains hidden, inaccessible at the heart of the galaxy of observables that revolve around it, as illustrated in Fig. 15 of Sect. 4.

To address all aspects related to the development of the concept of nuclear shape from its origins to the present day, and to better understand the significance of the recent advent of high-energy measurements in the trajectory of this concept, it is proposed to divide this review into three parts. In the first part, we reintroduce the minimal necessary formalism to make the discussion on the concept of nuclear shape more precise, and especially to properly introduce an essential concept for the subsequent discussion: nuclear translucency. The second part constitutes the core of this contribu-

¹ This is actually a remarkable point, and Mackintosh in his review notes “The existence of molecules like ethane, which are not naturally thought of as spherical, as a precedent for non-spherical quantum systems is not reflected in the nuclear physics literature.” [8].

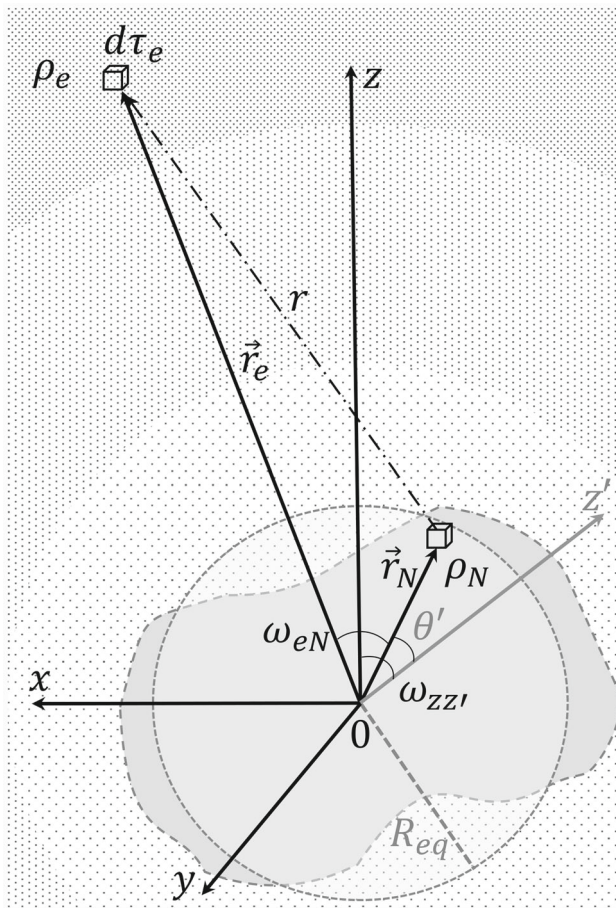


Fig. 1 A nuclear charge distribution ρ_N of arbitrary shape with an equivalent radius R_{eq} immersed in the electric charge distribution ρ_e of the electron cloud. The laboratory reference frame is denoted by the (x, y, z) axis system. Only the $(0z')$ main axis of the body-fixed intrinsic reference frame is shown. r is the distance between an elementary volume in the nuclear charge distribution located at $\vec{r}_N$ and an elementary volume in the electronic charge distribution located at $\vec{r}_e$

tion, focusing on the genesis and evolution of the concept of nuclear deformation. We revisit the major milestones in the development and utilization of the nuclear shape concept, divided into three major periods. This genesis and subsequent evolution have been deeply intertwined with advancements in techniques and the precision of experiments, aspects that will also be addressed. The final part, serving as a conclusion and outlook, will highlight the impact and historical significance of the advent of relativistic heavy-ion collisions in the context of nuclear shape.

2 The concept of nuclear shape: a few formal remarks

Before delving further into the history of the nuclear shape concept, it is necessary to recall some definitions and foundational ideas associated with this concept. And even before

even discussing nuclear shape, it is necessary to recall that at the root of everything lies the saturation of nuclear forces, their truly primary characteristic, at the core of all fundamental properties of the atomic nucleus, including its finite size. Let's call this "size", this spatial extension, R , a sort of radius that we will define better later. Without saturation of the nuclear forces, R would have no dependence on mass number A , being the radius of the collapsed state [9], of the order of the range of the interaction. It was however soon realized [10] that the volume of a nucleus is approximately proportional to A , so that the volume per nucleon is about the same in every nucleus, so that R scales with $A^{1/3}$ like $R = r_0 A^{1/3}$ where r_0 is a parameter practically constant over the entire nuclear chart: $r_0 = 1.2 - 1.4$ fm depending on the way the nuclear size is probed. This r_0 value has early been considered "reassuring," as it is of the same order of magnitude as the wavelength $\lambda = \frac{h}{\sqrt{2MT}}$ associated with a mass M proton or neutron possessing the expected average kinetic energy T of a dozen of MeV. As most bound nucleons within nuclei were expected to be in states possessing several nodes, this observed size was consistent with the idea that nuclei were composed of "packed" protons and neutrons, but also with the idea that the nuclear density spatial distribution may possess mass and charge irregularities at nucleon loci (not to mention non-locality effects).

2.1 Nuclear mass and charge distributions

So, at this stage of the story, it is evident that no one could have had any a priori idea about the shape that the spatial distribution of mass and charge must have taken to lead to the observed nuclear dimensions. Any distribution, even very complicated, showing some "clumps" at the locations of the nucleons, could have been imagined. However, bearing in mind that for most measurements, only time-averaged quantities could be attained, and considering that unless compelled by observations, one could settle for forms that were practical from a computational standpoint, pioneers proposed to adopt the most general form of the nuclear density distribution as [11]:

$$\rho_N(r) = \rho_0 f(r/B) \quad (1)$$

where ρ_0 is the central density of the charge or mass, $f(0) = 1$, and B is the "range" parameter, characterizing the size of the nucleus. The completely general situation is the one where the function $f(x)$ itself depends on angle but, initially (and probably even now), there was nothing in the observations that necessitated maintaining such generality and it was supposed (and it is still the case in contemporary applications) that the functional form of f does not vary with angle. The angular variations of the nuclear density distributions are then rather described by representing the range parameter B

as a function of θ' and φ' , the polar and azimuthal angles measured with respect to reference axes *attached* to the nuclear surface: the so called *intrinsic* frame, or *body-fixed* frame (subsequently, all quantities defined in this reference frame are primed, while those defined in the laboratory reference frame are not). In that framework, Eq. 1 should be regarded as the density distribution along a radius vector at some orientation relative to nuclear body-fixed axes, the radial shape is the same at each orientation, only the “range” varies.

The range parameter B of Eq. 1 has a different meaning for each different functional form of $f(x)$. To be able to define unambiguously a nuclear volume, and, even more interesting, a nuclear radius, the only way is to assume that the nuclear volume, regardless of its angular shape, is bounded by a sharp edge delineating a uniform distribution. This corresponds to using the simplest of radial shapes:

$$\begin{aligned} f(x) &= 1, \quad x \leq 1 \\ f(x) &= 0, \quad x > 1 \end{aligned} \quad (2)$$

It is sometimes forgotten that this assumption of a uniform distribution is underlying the vast majority of applications related to determining nuclear shape from experimental data. It was certainly clear from the beginning, as it was soon largely proven from electron scattering data [12], that the principles of quantum mechanics require the nuclear surface to be represented by a continuous function of moderate gradient and not by an abrupt drop to zero. But, with the radial shape described by Eq. 2 $B = R$ is an unambiguously defined nuclear radius when the distribution is spherically symmetric. Bodmer [13] first introduced the concept of “equivalent” radius, R_{eq} applicable to any non uniform shape: it is the radius of a spherical uniform distribution which gives the same mean square radius $\langle r^2 \rangle$ as the real distribution, where $\langle r^2 \rangle$ is related to Eq. 1 by²:

$$\langle r^2 \rangle = B^2 \frac{\int_0^\infty f(x) x^4 dx}{\int_0^\infty f(x) x^2 dx}$$

and one has $R_{eq} = \sqrt{\frac{3}{5} \langle r^2 \rangle}$.

These reminders, which may seem trivial at first glance, are here to remind us that it is appropriate to distinguish three properties of nuclear mass (or charge) distributions:

1. the size of the mass (or charge) distribution of the nucleus that is to say its “radius”;
2. the *radial* variation of the mass (or charge) distribution of the nucleus, commonly referred to in the literature as its “radial shape”;

3. the *angular* variation of the nuclear mass (or charge) distribution, often called the “angular shape” of the nucleus (or equivalently but rather rarely “geometrical” shape).

Actually, the link between the true subject of this article, the concept of nuclear geometrical or angular shape of atomic nuclei, and the issue related to nuclear surface is essential. Almost all applications related to the study of nuclear angular shapes involve completely disregarding the radial shape, by reducing it to a simple abrupt transition from central density to zero at the nuclear surface, that is, to a sharp boundary (the uniform radial distribution assumption of Eq. 2). Conversely, when analyzing measurements sensitive to the radial distribution of nuclear density, such as elastic electron scattering experiments at several hundreds MeV, which are sensitive to the internal charge distribution, efforts have been made to carefully avoid the issue of angular shape. Most of previous reviews on the subject, since the first *direct* determination of nuclear charge radii from electron scattering measurements in 1951 [14], focused on items 1 and 2 above, with the connection to item 3 often cautiously confined to a small subsection. However, it is important to bear in mind that the properties related to these three items cannot truly be measured separately. Ford and Hill in their early review [11] remind us that: “None of these quantities, of course, is measured alone. Most experiments can be interpreted in terms of certain weighted averages over the charge distribution—e.g., a mean square radius instead of a radius alone, or a radial moment instead of a radial shape, or a quadrupole moment instead of an angular shape. But, ideally, the picture one seeks is a successively more detailed description in terms of the size, the deviations of the radial variation from zero (corresponding to a constant density distribution), and the deviation of the angular variation from spherical symmetry”.

At that point, it is important also to recall that everything we know, even today, with some precision about geometrical shape of nuclear surfaces or even the size of nuclei, relies on what we know about their *charge distribution*, especially as we move away from the stability valley. There are both practical experimental reasons for this (charged probe particle, including quasi-point-like ones such as electrons, have been available since the early 1950s—and actually *s* electrons of the atomic cloud penetrating inside the nuclear interior, since much longer) and a fundamental reason: our ability to describe electromagnetic interactions is far more precise than our understanding of nuclear interactions.

It should be emphasized that the spatial distributions of protons and neutrons cannot be expected to be exactly identical, although there are compelling reasons to believe they are similar. Furthermore, the charge distribution is distinct from the proton distribution, representing a more complex entity that includes internal charge and spin current distributions of protons and neutrons. A detailed discussion of

² This definition was chosen because the first low-energy electron scattering experiments were measuring $\langle r^2 \rangle$.

these intricacies is beyond the scope of this article, but it is an important topic that has historically been addressed through the results of muonic X-ray spectra measurements in medium- to heavy-mass nuclei, and high-precision elastic electron scattering experiments [15]. More recently, this topic has been the subject of careful theoretical investigations within the framework of self-consistent mean-field calculations [16], in the context of the accumulating measurements of isotopic changes in mean-square charge radii $\delta\langle r_c^2 \rangle$ obtained from the precise experimental determination of isotope shifts of atomic hyperfine spectra (the importance of this latter observable, the isotope shift-remarkably, its measurement has been extended to a vast portion of the nuclear chart over the past three decades, see Fig. 1.2 in Ref. [17]—in the history of the nuclear shape concept will be highlighted later in this article.).

The nuclear density ρ_N , and notably the *deformed* nuclear density, has been historically parameterized in a variety of ways which correspond to different structures of the nuclear surface. It is customary to rewrite Eq. 1 in the form of an expansion in spherical harmonics in the intrinsic reference frame:

$$\rho_N(\mathbf{r}') = \sum_{\lambda\mu} \rho_{\lambda\mu}(r) Y_{\lambda\mu}(\theta', \varphi') \quad (3)$$

where $Y_{\lambda\mu}$ are spherical harmonics with, as usual, $-\lambda \leq \mu \leq +\lambda$ representing the $2\lambda + 1$ modes of deformation of order λ . If one assumes a uniform distribution bounded by a sharp-edged surface (the radial form of Eq. 2), then this surface can also be (in the body-fixed reference frame) expanded in spherical harmonics [18]:

$$R(\theta', \varphi') = R_0 \left[1 + \sum_{\lambda\mu} \alpha_{\lambda\mu} Y_{\lambda\mu}(\theta', \varphi') \right] \quad (4)$$

where R_0 is the radius of the sphere of same volume as the deformed shape (NB. for this reason, R_0 depends on the parameters defining the shape's surface. This will remain true whenever it appears in the equations throughout this article, even if this dependence is not explicitly stated). Inside of this surface lies a uniform charge density:

$$\rho_N(r) = \begin{cases} \frac{3Z}{4\pi R_0^3} & r \leq R(\theta', \varphi') \\ 0 & r > R(\theta', \varphi') \end{cases} \quad (5)$$

One often finds in the literature (especially in monographs) the notion that the first term truly relevant for the description of a nuclear shape is the one of order $\lambda = 2$ (quadrupole deformation) because $\lambda = 1$ corresponds to a translation of the nucleus. And, in general, most authors

agreed that odd-order deformation modes were expected to play only a minor role, as the associated shapes were reflection-asymmetric and would therefore lead to nuclear (intrinsic) states lacking parity as a good quantum number. However, V.M. Strutinsky [19, 20], A. Bohr and B. Mottelson [21, 22] pointed out a long time ago that an effective dipolar deformation component is not impossible but rather arises from the superposition of quadrupole and octupole modes.³

Restricting the expansion of Eq. 4 to $\lambda = 2$:

$$R(\theta', \varphi') = R_0 \left[1 + \sum_{\mu} \alpha_{2\mu} Y_{2\mu}(\theta', \varphi') \right] \quad (6)$$

the shape's description requires only two parameters, written, following the historical publication of A. Bohr [26] as β and γ :

$$\begin{aligned} \alpha_{20} &= \beta \cos \gamma \\ \alpha_{21} &= \alpha_{2-1} = 0 \\ \alpha_{22} &= \alpha_{2-2} = \frac{\beta}{\sqrt{2}} \sin \gamma \end{aligned} \quad (7)$$

where β measures the deviation from sphericity and γ from axiality. A *prolate* spheroid is obtained when $\gamma = 0$ and an *oblate* one when $\gamma = \pi/3$. Intermediate γ values describe ellipsoidal shapes without axial symmetry around any of the x' , y' , z' intrinsic-frame axes: with each increase of γ by $\pi/3$ the sequence of ellipsoidal deformations is repeated for a different orientation of the ellipsoidal forms relative the intrinsic-frame axes.

Quadrupole deformation modes play a particular role in the realm of nuclear shape. They are the first to be explicitly considered in models and often remain the sole modes, primarily starting with the axial mode, for simplicity. Therefore, it is useful to define already here some key (electric) quadrupole quantities involved in measurements.

2.2 The nuclear quadrupole moment: from spectroscopic to intrinsic

2.2.1 Definition of the quadrupole moment

The shape represented by Eq. 6 is the one envisioned at first order when studying nuclear manifestations in the laboratory frame that indicate a deviation from sphericity in a hypothetical intrinsic frame. As we will see in detail in Sect.

³ This topic was subsequently explored extensively, particularly in the context of the discovery of strong electric dipole (E1) strength in the spectroscopy of certain nuclei in the actinide region. Interested readers may refer to (among others) the work of Leander et al. [23], the excellent review by Butler and Nazarewicz [24] (and references therein), and the work of Åberg et al. for theoretical aspects [25].

3.1.1, the quantity that has historically revealed this manifestation (at least for *charge* distributions) is the *quadrupole moment*, and the discovery of its existence played a foundational role in the development of the concept of nuclear shape. While, as reminded in the Introduction, there is no operational definition of any parameter that would serve as a measure of nuclear shape, such a definition *does* exist for the nuclear quadrupole moment. The nuclear quadrupole moment was originally [27,28] introduced in atomic spectroscopy to account for deviations from the Landé rule observed in the hyperfine spectra of certain rare earth elements (see Sect. 3.1.1). Its operational definition, as well as its common name—spectroscopic quadrupole moment, Q_s —derive from this context. The measurement of this quantity is, of course, performed in the laboratory frame. However, as we will now see, it is possible to infer something from this laboratory-frame quantity about the deviation from sphericity in the nuclear charge distribution as envisioned in its intrinsic frame. It is important, nevertheless, to stress immediately that this is only feasible under specific conditions and requires the adoption of several drastic assumptions. The spectroscopic quadrupole moment is essentially a multiplicative factor in the energy term describing the interaction between the nuclear charge distribution and the electric field gradient generated by the surrounding electron cloud, once the nucleus is no longer treated as a point charge. It naturally arises in the series expansion of this energy term, enabled by the fact that the nuclear size is considerably smaller compared to the typical extent of the electron cloud. Considering, as illustrated in Fig. 1, a nuclear charge distribution $e\rho_N$ of arbitrary shape with an equivalent radius R_{eq} immersed in the electric charge distribution $-e\rho_e$ of the electron cloud, this corresponds to the condition where $r_e \gg r_N$ on the average. Following Hendrik Casimir⁴ [28], who was the first to derive this term of electrostatic energy and to isolate (thus define) the spectroscopic quadrupole moment, we consider the difference between the true electrostatic interaction and the interaction between the electrons and a point charge.

$$\Delta V_{el} = e \int \rho_N \varphi_e d\tau - \frac{Ze^2}{4\pi\epsilon_0} \int \frac{\rho_e}{r} d\tau$$

where φ_e represents electrostatic potential created by the electron cloud (a function of x , y and z) and $d\tau$ is an elementary volume. ΔV_{el} is evaluated in the laboratory reference frame (x , y , z) whose origin (0) can be taken as the center of gravity both of the nuclear and electron charge

distributions. For electrons that have no charge distribution within the nuclear volume itself—that is, all but the s and $p_{1/2}$ electrons—it is possible to consider that $r_e \gg r_N$ on the average and interesting to expand φ_e in powers of x , y , z around the origin, leading to the following form:

$$\Delta V_{el} = e \int \rho_N \varphi(0) d\tau \quad (8)$$

$$+ e \int \rho_N \left[x \frac{\partial \varphi_e}{\partial x}(0) + y \frac{\partial \varphi_e}{\partial y}(0) + z \frac{\partial \varphi_e}{\partial z}(0) \right] d\tau \\ + \frac{1}{2} e \int \rho_N \left[x^2 \frac{\partial^2 \varphi_e}{\partial x^2}(0) + xy \frac{\partial^2 \varphi_e}{\partial x \partial y}(0) + \dots + z^2 \frac{\partial^2 \varphi_e}{\partial z^2}(0) \right] d\tau \\ - Z \frac{e^2}{4\pi\epsilon_0} \int \frac{\rho_e}{r} d\tau \quad (9)$$

The first (monopole) and last terms in this equation cancel each other. The second term represents the interaction between a nuclear electric dipole moment and the electric field created by the electronic cloud at the origin. It is zero because a non-degenerate stationary state of a quantum system has a definite parity.⁵ Strictly speaking this would hold true for the ground states of nuclei if their center of gravity and center of charge were exactly coincident, which they are not (as mentioned earlier), as evidenced by the presence of significant $E1$ strength in some of them. However, the effect is sufficiently small in an atomic physics context [30] to be neglected for the purposes of this discussion. In the third term, let's call it ΔV_Q , one recognizes a scalar product of the tensor of the gradient of the electric field created by the electrons at the origin:

$$\begin{pmatrix} \frac{\partial^2 \varphi_e}{\partial x^2} & \frac{\partial^2 \varphi_e}{\partial x \partial y} & \frac{\partial^2 \varphi_e}{\partial x \partial z} \\ \frac{\partial^2 \varphi_e}{\partial y \partial x} & \frac{\partial^2 \varphi_e}{\partial y^2} & \frac{\partial^2 \varphi_e}{\partial y \partial z} \\ \frac{\partial^2 \varphi_e}{\partial z \partial x} & \frac{\partial^2 \varphi_e}{\partial z \partial y} & \frac{\partial^2 \varphi_e}{\partial z^2} \end{pmatrix}_0 \quad (10)$$

and another tensor, which is interpreted as the tensor of the quadrupole moment (in Cartesian coordinates):

$$\begin{pmatrix} \int \rho_N x^2 d\tau & \int \rho_N xy d\tau & \int \rho_N xz d\tau \\ \int \rho_N yx d\tau & \int \rho_N y^2 d\tau & \int \rho_N yz d\tau \\ \int \rho_N zx d\tau & \int \rho_N zy d\tau & \int \rho_N z^2 d\tau \end{pmatrix} = \begin{pmatrix} Q_{xx} & Q_{xy} & Q_{xz} \\ Q_{yx} & Q_{yy} & Q_{yz} \\ Q_{zx} & Q_{zy} & Q_{zz} \end{pmatrix} \quad (11)$$

To recover the definition of Casimir's spectroscopic quadrupole moment, Eq. 10 must be simplified by assuming that the electric field created by the electrons at the nucleus, $\mathcal{E}$, possesses cylindrical symmetry along the (0 z)-axis, so that the off-diagonal terms of the tensor of Eq. 10 cancel and:

⁴ We here follow the essential steps of Casimir's derivation in a version (and formalism) very close to that of Kopfermann [29]. Hans Kopfermann is the author of the foundational monograph on the science of nuclear moments, which has inspired all subsequent generations of physicists in this field (including the author). The novice reader in this subject would benefit greatly from referring to this work.

⁵ An elegant demonstration can be found in Ref. [9], p. 24.

$$\Delta V_Q = \frac{1}{2}e \left[Q_{xx} \left(\frac{\partial^2 \varphi_e}{\partial x^2} \right)_0 + Q_{yy} \left(\frac{\partial^2 \varphi_e}{\partial y^2} \right)_0 + Q_{zz} \left(\frac{\partial^2 \varphi_e}{\partial z^2} \right)_0 \right] \quad (12)$$

Since we have already assumed that the electronic charge density inside the nuclear volume is either zero or negligible (absence of spherical s and $p_{1/2}$ electron distributions), we have $\nabla^2 \varphi_e = 0$ around the origin. It follows from this and the cylindrical symmetry of $\mathcal{E}$ around the $(0z)$ -axis that:

$$\left(\frac{\partial^2 \varphi_e}{\partial z^2} \right)_0 = -2 \left(\frac{\partial^2 \varphi_e}{\partial x^2} \right)_0 = -2 \left(\frac{\partial^2 \varphi_e}{\partial y^2} \right)_0 \quad (13)$$

Using this relationship and Eq. 11, Eq. 12 may be rewritten as:

$$\Delta V_Q = \frac{1}{4}e \left(\frac{\partial^2 \varphi_e}{\partial z^2} \right)_0 \int (2z^2 - x^2 - y^2) \rho_N(x, y, z) d\tau_N \quad (14)$$

or

$$\Delta V_Q = \frac{1}{4}e \left(\frac{\partial^2 \varphi_e}{\partial z^2} \right)_0 \int (3z^2 - r_N^2) \rho_N(x, y, z) d\tau_N \quad (15)$$

It is the integral (running over the nuclear volume) which occurs in Eq. 15 which serves as the basis for the definition of the *spectroscopic* quadrupole moment of the nuclear charge distribution $e\rho_N$:

$$eQ_s \equiv e \int (3z^2 - r_N^2) \rho_N d\tau_N \quad (16)$$

If $\omega_{zz'}$ be the relative angle between the main axes $(0z)$ and $(0z')$ of the laboratory and intrinsic frames, as shown in Fig. 1, then one can show that:

$$\begin{aligned} & \int (3z^2 - r_N^2) \rho_N d\tau_N \\ &= \frac{1}{2} (3 \cos^2 \omega_{zz'} - 1) \int (3z'^2 - r_N^2) \rho_N d\tau'_N \end{aligned} \quad (17)$$

then

$$e \int (3z'^2 - r_N^2) \rho_N d\tau'_N \equiv eQ_0 \quad (18)$$

is the definition of the *intrinsic* quadrupole moment.

The concept of electric quadrupole moment can be generalized by introducing a multipole moment tensor $q_{\lambda\mu}$ such as [31] (we drop the index N from now on as it is clear that this quantities are evaluated in the nuclear interior):

$$q_{\lambda\mu} = \int r^\lambda Y_{\lambda\mu}^*(\theta, \varphi) \rho(\mathbf{r}) d\tau \quad (19)$$

In nuclear physics, because axially symmetric quadrupole shapes have played (and still play) a privileged or reference role in the phenomenology of deformation, spherical coordinate notations are generally preferred. However, for completeness, we provide the expression of the quadrupole moment tensor in Cartesian coordinates [32]:

$$Q_{ij} = \int (3r_i r_j - r^2 \delta_{ij}) \rho(\mathbf{r}) d\tau \quad (20)$$

where r_i and r_j stand for (x, y, z) , and δ_{ij} is the Kronecker delta. It may be useful, for the sake of clarity, to remind that the relation between Cartesian and spherical moments is established through transformation equations. For instance, the Cartesian component Q_{zz} relates to the spherical component q_{20} as follows [33]:

$$q_{20} = \frac{1}{2}(2Q_{zz} - Q_{xx} - Q_{yy}) \quad (21)$$

Furthermore, off-diagonal elements such as Q_{xy} and Q_{xz} are related to spherical components q_{2m} for $m = \pm 1, \pm 2$, through combinations like [34]:

$$q_{21} = -Q_{xz} + iQ_{yz} \quad (22)$$

or

$$q_{22} = Q_{xx} - Q_{yy} + 2iQ_{xy} \quad (23)$$

But in general, what is commonly referred to in the literature [8] as the basic quantity for defining multipole moments is:

$$Q_\lambda \equiv \int r^\lambda Y_{\lambda 0}(\theta, \varphi) \rho(\mathbf{r}) d\tau \quad (24)$$

that is no more than the one from Eq. 19 restricted to axial modes $\mu = 0$.

It is important to emphasize that, in order to arrive at these operational definitions (i.e., that ΔV_Q is indeed a measurable quantity in hyperfine spectra within the laboratory frame) of both the spectroscopic and intrinsic quadrupole moments, we have relied on two main assumptions (which more recent literature seldom reiterates): the electron cloud generates a field with axial symmetry around an axis, chosen as the principal axis $(0z)$ of the laboratory frame, and this field is assumed to be created by charges distant from the nucleus, meaning that we exclude the s and $p_{1/2}$ electrons⁶. Up until now, we have also deliberately ignored the fact that the nucleus is a quantum object and that its total angular momentum I is a good quantum number. It is now important to recall this, as the orientation of this system in space can only be defined in relation to the orientation of its total angular momentum. Historically,

⁶ This point, seemingly entirely forgotten in recent discussions, raises some questions, given that atomic transitions are carefully selected for their s and $p_{1/2}$ content, which provide good sensitivity to isotopic volume shifts associated to significant isotope displacements of the centroid of the hyperfine spectra, enabling the measurement of the mean square charge radius variation between isotopes in experiments.

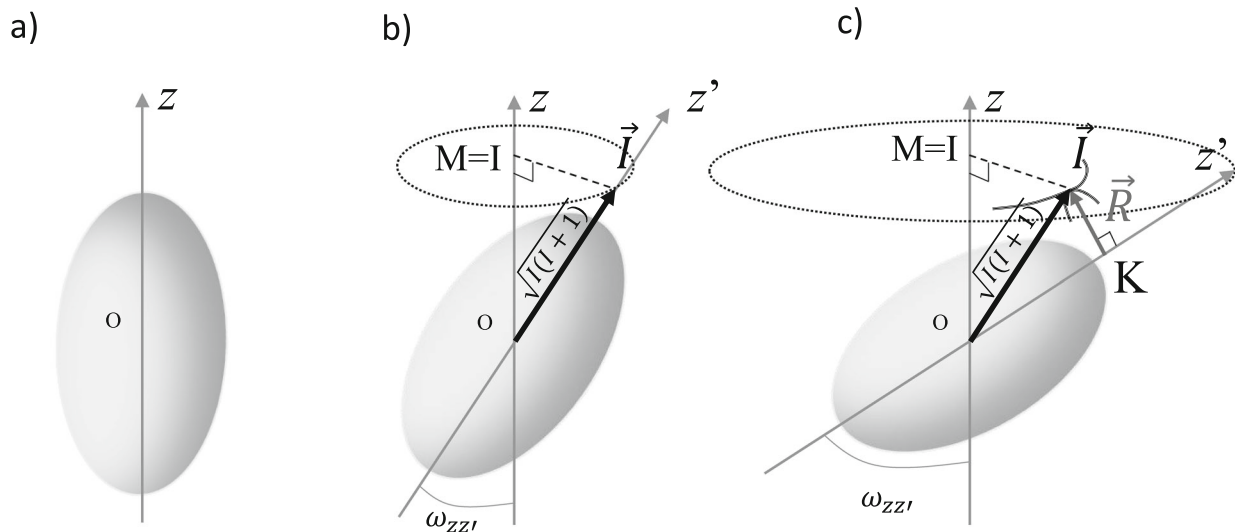


Fig. 2 Orientation in space of the main axis ($0z'$) of the body-fixed reference frame in the hypothesis of a nucleus with of revolution ellipsoid shape. **a** Classical case, the laboratory and intrinsic axes systems are identical. **b** Situation where the total angular momentum of the nucleus

is oriented along the symmetry axis of the revolution ellipsoid. **c** Situation for a rotational state with total angular momentum I and rotation angular momentum R

these quantum mechanical considerations immediately led to a refinement of the definition of the spectroscopic quadrupole moment by specifying that the nuclear charge distribution is evaluated in the state of maximum projection $M = I$ along the laboratory-frame axis ($0z$). If (following e.g. [9]) one assumes that the probability of finding the i th nucleon in the nuclear volume element $d\tau$ around the position $\mathbf{r}$ is given by $P_i d\tau$ where:

$$P_i(\mathbf{r}) = \int |\Psi_{IM}(\mathbf{r}_1, \dots, \mathbf{r}_A)|^2 d\tau_1 \dots d\tau_{i-1} d\tau_{i+1} \dots d\tau_A \quad (25)$$

$\Psi_{IM}(\mathbf{r})$ being the wave function of the nucleus with total angular momentum I and ($0z$)-projection number M , then the charge density $\rho(\mathbf{r})$ can be replaced by a quantum equivalent:

$$e\rho(\mathbf{r}) = \sum_{i=1}^Z eP_i(\mathbf{r}) \quad (26)$$

and

$$eQ_s = \sum_{i=1}^Z \int e(3z_i^2 - r_i^2) |\Psi_{II}(\mathbf{r}_1, \dots, \mathbf{r}_A)|^2 d\tau \quad (27)$$

(actually, to be more precise, it is a time average value that accounts for the precession of the angular momentum around the ($0z$)-axis, see panel (b) in Fig. 2). The reason for this choice is that it is in this state that the direction of ($0z'$)

is most clearly defined in the laboratory frame. The projection over the ($0z$)-axis can never be complete because the norm $\sqrt{I(I+1)}$ of the angular momentum vector is obtained using $\langle \mathbf{I}^2 \rangle = I(I+1)\hbar^2$, while its projection onto the quantization axis for a given magnetic substate $|IM\rangle$ is $\langle I_z \rangle = M\hbar$. Considering Fig. 2(b), it is straightforward to understand that the relative angle between the axes ($0z$) and ($0z'$) is given by:

$$\cos \omega_{zz'} = \frac{M}{\sqrt{I(I+1)}} \quad (28)$$

and the maximum value of $\cos \omega_{zz'}$ is $\frac{I}{\sqrt{I(I+1)}}$. The classical limit is recovered for $I \rightarrow \infty$, $\cos \omega_{zz} \rightarrow 1$, which corresponds to the situation represented in panel (a) of Fig. 2, where the entire quadrupole distribution is expressed in the laboratory frame (an impossible measurement in a quantum system, of course), and the ($0z$) and ($0z'$) axes become indistinguishable. One can note, for completeness, that for consistency with Eq. 27, an alternative form of the expression for the multipole moment operator (Eq. 24) can be written as:

$$\begin{aligned} eQ_\lambda &= e \sum_{i=1}^Z r_i^\lambda Y_{\lambda 0}(\theta_i, \varphi_i) \\ &= e \left[\frac{(2\lambda + 1)}{4\pi} \right]^{1/2} \sum_{i=1}^Z r_i^\lambda P_\lambda(\theta_i) \end{aligned} \quad (29)$$

where $P_\lambda(\theta)$ is the Legendre polynomial of order λ . One then has:

$$eQ_s = e\sqrt{16\pi/5}\langle Q_2 \rangle_{II} \quad (30)$$

It is useful, for the upcoming discussion on the origins of the nuclear shape concept and its (subtle yet essential, as we shall see) connection with the emergence of the spherical shell model, to evaluate Eq. 30 in the case of a single-particle state $|j, m_j\rangle$ of this model. It suffices to evaluate $\langle Q_2 \rangle_{II}$ in this equation for a pure single-particle state, i.e., $I = j$ and $M = m_j = j$ (maximum projection of $\mathbf{j}$ along the laboratory frame ($0z$) axis), which gives [29]:

$$\langle Q_2 \rangle_{II} = \langle NI | r^2 | NI \rangle \left\langle l \frac{1}{2} j j | Y_{20} | l \frac{1}{2} j j \right\rangle \quad (31)$$

leading to:

$$eQ_s(jj) = -e \frac{2j-1}{2j+2} \langle NI | r^2 | NI \rangle \quad (32)$$

The value for the quadrupole moment in other projection m_j -substates are obtained from $Q_s(jj)$ via [29]:

$$Q_s(jm_j) = Q_s(jj) \cdot \frac{3m_j^2 - j(j+1)}{j(2j-1)} \quad (33)$$

It is then immediately evident that each magnetic substate $|jm_j\rangle$ of a single-particle state $|l_j\rangle$ in the spherical potential of the shell model—except, of course, for $j = 1/2$ —has a nonzero quadrupole moment in the laboratory frame. In this sense, each of these substates can be considered to possess a *shape*. It is also evident that the spectroscopic quadrupole moment of a nucleus with total angular momentum $I = j$, just beyond a double shell closure (which would be described as a doubly magic core plus one particle), is predicted to be negative. This is consistent with the intuitive picture of these magnetic substates as *orbitals*—even though the term is nowadays used interchangeably for all types of single-particle states: the semiclassical trajectory of the orbit lies predominantly in a plane orthogonal to the angular momentum j , which itself precesses around the ($0z$)-axis in the state $|j, m_j = j\rangle$, resulting in a sort of equatorial “bulge,” i.e., an oblate shape. Equation 33 also illustrates that one has $Q_s(jm_j) = Q_s(j - m_j)$ and that $\sum_{-j}^{+j} Q_s(jm_j) = 0$ (a closed j -shell is spherical). The general result for a nuclear state of good angular momentum I formed by the coupling of ζ identical particles filling a $(2j+1)$ -degenerate j -shell is [35]:

$$Q_s(I = j) = Q_s(jj) \frac{2j+1-2\zeta}{2j-1} \quad \text{for } \zeta \text{ odd} \quad (34)$$

and

$$Q_s(I = 0) = 0 \quad \text{for } \zeta \text{ even} \quad (35)$$

$Q_s(jj)$ being negative, as noted earlier, one has negative $Q_s(I = j)$ for a less than half-filled j -subshell, $\zeta < \frac{2j+1}{2}$ and positive for a less than half-empty j -subshell, $\zeta > \frac{2j+1}{2}$. A single hole in a filled j -subshell corresponds to $\zeta = 2j$, and it comes from Eq. 34 that $Q_s(\text{single particle}) = -Q_s(\text{single hole})$. This result is just as intuitive as before: a hole in a closed and therefore spherical j -subshell corresponds to the absence of the “bulge” created by the previously mentioned orbit and thus appears as a spherical distribution with a superimposed bulge of opposite charge. Incidentally, at exactly mid-shell, since the number of particles and holes is equal, the positive and negative bulges cancel each other out, and the quadrupole moment vanishes.

At this point, an additional conceptual difficulty arises: the situation shown in panel (b) of Fig. 2 where the direction in which the total angular momentum vector of the nucleus points coincides with the intrinsic axis ($0z'$), which also serves as the principal axis—i.e., the symmetry axis of the ellipsoid as previously discussed—is, by definition, impossible within the framework of the rotational model (which we will discuss later), where the total angular momentum of the nucleus is considered to originate from the rotation of its shape.⁷ This image originates from Casimir’s original demonstration [27,28], which aimed to simplify the calculation of the electrostatic interaction term ΔV_Q between a nucleus with total angular momentum I and an electron cloud with total angular momentum J . Casimir assumed that the charge distribution of the nucleus exhibited axial symmetry around the axis defined by I (i.e., the axis ($0z'$) in panel b of Fig. 2), while the electronic electric field displayed axial symmetry around the axis defined by J (i.e., the axis ($0z$) in panel b of Fig. 2)). The energy term can then be explicitly expressed as a function of the operators $\mathbf{I}$ and $\mathbf{J}$ as [36]:

$$\Delta V_Q = \frac{eQ_s \langle (\partial^2 \varphi_e / \partial z^2)_0 \rangle_{JJ}}{2I(2I-1)J(2J-1)} \times \left[3(\mathbf{I} \cdot \mathbf{J})^2 + \frac{3}{2} \mathbf{I} \cdot \mathbf{J} - \mathbf{I}^2 \mathbf{J}^2 \right] \quad (36)$$

When spectroscopic quadrupole moments are extracted from the measurement of the term ΔV_Q in hyperfine spectra⁸, this

⁷ In quantum mechanics, rotation around the symmetry axis is just not observable, as it does not correspond to any measurable change in the system’s state.

⁸ It is common to refer to the term $B = eQ \langle (\partial^2 \varphi_e / \partial z^2)_0 \rangle_{JJ}$ as the “quadrupole coupling constant”. The lifting of the degeneracy of atomic fine-structure levels originates from this quadrupole term added to a magnetic term (see e.g. Ref. [29]) that must also be determined in order to isolate ΔV_Q from the measurement of the hyperfine spectrum. Furthermore, according to Eq. 36, in order to extract Q_s from ΔV_Q , it is necessary to know the electric field gradient generated by the electron cloud in the region of the nucleus $\langle (\partial^2 \varphi_e / \partial z^2)_0 \rangle_{JJ}$. The most reliable Q_s values come from measured B values in atoms or small molecules, combined with precise *ab initio* calculations of the electric field gradients

expression is systematically applied even though the assumptions underpinning its derivation are rarely reiterated.

In the framework of the rotational model, a more accurate depiction corresponds to panel c of Fig. 2 (inspired by Fig. 23 from Ref. [40]), where the angular momentum of collective rotation is perpendicular to the symmetry axis ($0z'$), and K is defined as the quantum number corresponding to the projection of I on this axis. Similarly to the ($0z$)-axis, alignment with the ($0z'$)-axis can therefore never be perfect.

In this case, the relationship between the spectroscopic and intrinsic quadrupole moments is established by these relations [41]:

$$Q_s = \langle IK20|IK\rangle\langle II20|II\rangle Q_0 \quad (37)$$

$$= \frac{3K^2 - I(I+1)}{(I+1)(2I+3)} Q_0 \quad (38)$$

where the two Clebsch-Gordan coefficients in Eq. 37 have actually an intuitive geometric interpretation. Indeed, considering the drawing in panel (c) of Fig. 2, we see that the total angular momentum vector of the nucleus, $\mathbf{I}$, must simultaneously undergo precession around both axes ($0z$) and ($0z'$). This can only occur along the small arc segments drawn at the tip of the vector $\mathbf{R}$ and at the end of the dashed line representing its projection on the ($0z$) axis because I , M , and K must remain constant at all times. These coefficients, in a semi-classical picture, thus measure the portions of the two precession circles allowing the fulfillment of all these conditions. This double precession motion with constant I , M , and K can only be achieved if the axis ($0z'$) itself also precesses. This precession is represented by the large dashed ellipse and maintains the $\omega_{zz'}$ angle constant, as if the two precession circles around the ($0z$) and ($0z'$) axes rotated against each other like the interlocking gears of a mechanical angular transfer system. One should note that, as a direct consequence of Eq. 38, the sign of the spectroscopic quadrupole moment Q_s cannot tell us if the nucleus is prolate or oblate. For states with total angular momentum I high enough in comparison with K the spectroscopic quadrupole moment is negative for a nucleus with prolate deformation i.e. a positive intrinsic quadrupole moment Q_0 .

It should also be perfectly clear from Eq. 38 that Q_s , whatever the circumstances, can only ever be less in absolute value

than Q_0 . This reduction is caused by the lack of alignment of the nuclear symmetry axis with angular momentum vector, and the lack of alignment between the angular momentum vector and the laboratory ($0z$) axis. While Q_s can indeed be derived from a measured quantity ΔV_Q at the cost of a number of assumptions that we have discussed at length, Q_0 remains, on the other hand, entirely beyond the reach of direct measurement. As we will see later, we can only ensure the validity of the scheme depicted in Fig. 2(c) if the same value of Q_0 consistently produces different measured values in the laboratory frame. In all cases, as the quadrupole moment is an operator akin to a spherical harmonic of order 2, and thus analogous to an angular momentum $L = 2$, when evaluated in nuclear states with good angular momentum $I < 1$, its expectation value vanishes⁹ $\langle I = 0 | r^2 Y_{20} | I = 0 \rangle = 0$.

Aage Bohr was the first to propose a relationship between Q_s and Q_0 . As early as 1950 [42], he observed that for a nucleus in a $I = K$ ground state, the spectroscopic and intrinsic quadrupole moments are related by:

$$Q_s = \mathcal{P}_I \cdot Q_0 \quad (39)$$

where

$$\mathcal{P}_I = \frac{I(2I-1)}{(I+1)(2I+3)} \quad (40)$$

The projection factor $\mathcal{P}_I$ of Eq. 40 consistently vanishes for $I = 0$ or $1/2$. These equations also nicely illustrate the fact that Q_0 is the quantity that can be interpreted as a classical quadrupole moment, as $Q_s \rightarrow Q_0$ when $I \rightarrow \infty$ (Panel (a) of Fig. 2).

2.2.2 Connection between quadrupole moment and nuclear shape

Now that the concept of the quadrupole moment has been properly introduced and (operationally) defined, it is appropriate to investigate its connection with nuclear shape, which is far from unambiguous. Revisiting the charge distribution expressed as its expansion in spherical harmonics (Eq. 3), limiting this expansion to two terms:

$$\rho_N(\theta') = \rho_{00} + \rho_{20} Y_{20}(\theta') \quad (41)$$

(only the polar angle is necessary in this case) would correspond to a nucleus possessing only an electric quadrupole moment and no higher-order moments. However, in the case of a homogeneous charge distribution bounded by a sharp surface, no simple shape exists that confers solely an electric quadrupole moment without higher-order components. Löbner et al. [43], in their 1970 review, noted that this distinction

Footnote 8 continued

at the nucleus [37]. But, when dealing with unstable nuclei, the usual way rather consists in extracting the ratio of the quadrupole moments of a pair of different isotopes of the same element considering that the atomic structure remains the same, $B/B^{\text{ref}} \simeq Q_s/Q_s^{\text{ref}}$ [17,38] (for instance, B and Q_s being determined by laser spectroscopy while B^{ref} and Q_s^{ref} with other methods, e.g. Nuclear Magnetic Resonance, for a reference stable isotope). The scaling is only exact for a point nucleus, the uncertainty introduced on scaling depends on the differential change in the distribution of nuclear charge with neutron number between the two isotopes [39].

⁹ See the elegant, clear demonstration in Ref. [9], p. 26.

led to consider two different shapes for axially deformed prolate nuclei: the *revolution ellipsoid*, which possesses only a quadrupole moment, and the *quadrupoloid*, which exhibits a quadrupole deformation but does not imply it has exclusively a quadrupole moment. The definition of the revolution ellipsoid is specified by the length of the semi-axis c parallel to the $(0z')$ axis and semi-axis a perpendicular to the $(0z')$ direction. Blatt and Weisskopf [9] define for that case a deformation (or excentricity) parameter $\eta \equiv (c^2 - a^2)/(c^2 + a^2)$ and a mean square radius $R_{rms}^2 \equiv \frac{1}{2}(a^2 + c^2)$ so that the relation between the shape and the intrinsic quadrupole moment is simple [43]¹⁰:

$$Q_0 = \frac{4}{5} Z \eta R_{rms}^2 \quad (43)$$

However the expansion of this shape in spherical harmonics leads to [43]:

$$R(\theta') = a (\alpha_{00} Y_{00} + \alpha_{20} Y_{20} + \alpha_{40} Y_{40} + \dots), \quad (44)$$

where the coefficients $\alpha_{00}, \alpha_{20}, \alpha_{40} \dots$ have all finite values. For instance for a prolate deformation with $\beta \approx 0.3$, the coefficients take the following values:

$$R(\theta') = a (3.88021 \cdot Y_{00} + 0.33323 \cdot Y_{20} + 0.03701 \cdot Y_{40} + 0.0046 \cdot Y_{60} + \dots) \quad (45)$$

The quadrupoloid on the other hand is defined by limiting the development of Eq. 4 to the purely axial expression:

$$R(\theta') = R_0 (1 + \alpha_{00} Y_{00} + \alpha_{20} Y_{20}), \quad (46)$$

nota bene - α_{20} is then the usual Bohr axial deformation parameter β of Eq. 7 and the term $\alpha_{00} Y_{00}$ is needed for the constant-volume condition.

In that case one has:

$$Q_0 = \frac{3}{\sqrt{5\pi}} Z R_0^2 \beta \times \left(1 + \frac{1}{8} \sqrt{\frac{5}{\pi}} \beta + \frac{5}{8\pi} \beta^2 - \frac{1}{192} \cdot \frac{5}{\pi} \sqrt{\frac{5}{\pi}} \beta^3 + \dots \right) \quad (47)$$

It follows that after measuring ΔV_Q and extracting Q_s via Eq. 36, the only feasible way to interpret this quantity in terms of nuclear shape is to assume axial symmetry, apply the reference frame transformation formula (e.g. Eq. 39), and deduce a value for Q_0 . However, at this stage—unless an

additional measurement granting access to the nuclear hexadecapole moment is performed—it will remain impossible to determine whether the deformation parameters should be extracted using Eqs. 43 or 47.

This serves to illustrate one of Mackintosh's cautions [8], reminding us that the quadrupole moment no more defines the quadrupole shape than the root mean square radius defines the monopole shape. Various possible shapes with different parameterizations but the same expectation value for $r^2 Y_{20}$ correspond to different surface structures of the nucleus with different values of $r^n Y_{20}$ for $n \neq 2$.

The rotational ellipsoid and the quadrupoloid were initially adopted for the sake of computational simplicity but likely also because, at the time, there was no compelling reason to consider more complex shapes. They were therefore those envisioned at first order when studying nuclear manifestations in the laboratory frame that indicate a deviation from sphericity in a hypothetical intrinsic frame (typically for the derivation of Eq. 36). We now know, of course, that there is no reason to assume, except in rare cases, that any real nucleus would actually adopt such basic shapes. It then becomes clear that linking the measurement of a single quantity, such as Q_s , to the shape of the nucleus is no longer possible, as at least one additional parameter must be introduced—such as the asymmetry angle γ from Eq. 7—even when limiting the discussion to $\lambda = 2$ distributions. The most usual departure from axial symmetry corresponds to Eq. 3 limited to $\lambda = 2$ with $\rho_{22}(r) = \rho_{2-2}(r)$. A nucleus with non-zero ρ_{20} , $\rho_{2\pm 2}$ components has three reflection planes (hence is “triaxial”). The treatment of this shape as a rotating rigid body (within the rotational model that we will discuss a bit later) implies three different moments of inertia and has been first achieved by Davydov and Filippov [45]. In that context the intrinsic quadrupole moment tensor components are given by [46]:

$$\begin{aligned} q_{20}^{(\kappa)} &= Q_c \cos\left(\gamma - \frac{2\pi}{3}\kappa\right), \\ q_{22}^{(\kappa)} &= q_{2,-2}^{(\kappa)} = \frac{Q_c}{\sqrt{2}} \sin\left(\gamma - \frac{2\pi}{3}\kappa\right), \\ q_{21}^{(\kappa)} &= q_{2,-1}^{(\kappa)} = 0, \end{aligned} \quad (48)$$

where $\kappa = 1, 2, 3$ denotes the axis to which the projection μ is related, and

$$Q_c = \frac{3}{\sqrt{5\pi}} R_0^2 Z \beta$$

is the “core” intrinsic quadrupole moment.

Furthermore, for triaxial nuclei, the quantum number K is no longer a good quantum number, and the straightforward relationship between the intrinsic Q_0 and spectroscopic Q_s quadrupole moments is lost (in other words, Eq. 38 is no

¹⁰ One also finds in later literature [44]:

$$Q_0 = \frac{2}{5} Z R^2 \frac{(c/a)^2 - 1}{(c/a)^{2/3}} \quad (42)$$

where $R = 1.2 \cdot A^{1/3}$ fm.

longer valid). To appreciate the complexity, one can refer to Ref. [47], where Hecht and Satchler derive this relationship as a sum over the various K -components of the intrinsic wave function (Eq. 25 in that reference) in the case of odd nuclei.

To be thorough, it should be noted that there is only one trivial (but admittedly uninformative) case when the triaxial degree of freedom is relaxed. This is the case of completely “soft” or γ -unstable nuclei, where the shape adopts any angle γ with equal probability for a fixed β . The dynamics of this nuclear shape were treated by Wilets and Jean [48]. In this scenario, Q_0 vanishes, and the dynamics can no longer be associated with the rotation of a fixed shape.

It is therefore essential to retain two key points from this discussion on the quadrupole moment. The first is that Q_s , derived from measurements, is indeed a quantity sensitive to the shape of the nucleus but is insufficient on its own to fully specify it. Q_0 , on the other hand, is a purely theoretical quantity defined in a body-fixed reference frame. Its manifestation in observable properties, such as static or dynamic electromagnetic characteristics, involves a reference frame transformation, which is straightforward only under specific conditions (namely, an axially symmetric quadrupolar charge distribution). As we will see in the following paragraph, further insight can be gained by including the measurement of electric quadrupole transition matrix elements in the discussion.

2.3 Kumar’s quadrupole invariants

Considering, as we have just done ourselves, that the spectroscopic quadrupole moment of a nucleus vanishes if $I = 0$ or $\frac{1}{2}$ (for purely geometrical reasons), or if the nucleus has equal probability to be oblate or prolate (e.g. the Wilets-Jean limit [48], see above) without this implying that the nucleus has a spherical charge distribution, Krishna Kumar set out to find quantities that would not depend on any model but would still be able to give a measure of the intrinsic shape of the nucleus [49].

Using the basic operator of Eq. 19 with $\lambda = 2$, Kumar defines what he calls a n -body quadrupole moment operator:

$$Q^{(n)} = ([Q_2 \times Q_2 \cdots \times Q_2]_2 \cdot Q_2) \quad (49)$$

This is a scalar operator where the term in square brackets with subscript 2 signifies a tensor product series of rank 2, and the entire expression within parentheses denotes a scalar product. The definition of the Q_2 operators in this expression are very close to the standard definition of Eq. 29 (but with unspecified μ , and the elementary charge e is included in the operator):

$$Q_{2\mu} = e \sum_{i=1}^Z r_i^\lambda Y_{\lambda\mu}(\theta_i, \varphi_i) \quad (50)$$

Since $Q^{(n)}$ is defined as a scalar operator it can have non-vanishing matrix elements for any nuclear state, denoted hereafter by $k, l, m \cdots (|k \equiv k, I_k, \Pi_k\rangle, |l \equiv l, I_l, \Pi_l\rangle$ etc.). Kumar exploits only the diagonal matrix elements of his n -body operator, denoted by:

$$q_k^{(n)} \equiv \langle k, M_k | Q^{(n)} | k, M_k \rangle = (2I_k + 1)^{-1/2} \langle k | Q^{(n)} | k \rangle \quad (51)$$

The heart of Kumar’s method is to develop these n -body moments $q_k^{(n)}$ as sums of products of reduced matrix elements of the quadrupole operator, derived from $B(E2; k \rightarrow l) = M_{kl}^2 / (2I_k + 1)$ associated with specific transitions to other states $l, m \cdots$, the required $B(E2)$ values being directly obtainable from experimental data (see Fig. 3 for a pictorial illustration). As an example, the 2-body moment of a state k is given by:

$$\begin{aligned} q_k^{(2)} &= (2I_k + 1)^{-1} \sum_l M_{kl}^2 \\ &= \frac{5(I_k + 1)(2I_k + 3)}{16\pi I_k(2I_k - 1)} [Q_s(k)]^2 + \sum_{l \neq k} B(E2; k \rightarrow l) \end{aligned} \quad (52)$$

where the diagonal matrix element M_{kk} has been directly written under the form of $Q_s(k)$, the spectroscopic quadrupole moment of state k . Since this 2-body moment is a sum of squares of various $E2$ matrix elements it would vanish only if each of these matrix elements vanishes—not only the spectroscopic quadrupole moment but also all $E2$ transition matrix element connecting state k to any other state. Conversely $q_k^{(2)}$ does not vanish even in the case where $Q_s(k) = 0$ for purely geometrical reasons, i.e. for $I_k = 0$ or $\frac{1}{2}$, if the state k is connected to at least one other state by an $E2$ transition. Therefore this 2-body moment, $q_k^{(2)}$, can be interpreted as a completely general measure of the quadrupole deformation of the state k . It is insufficient however to determine if the nucleus is prolate or oblate, for this the 3-body moment:

$$\begin{aligned} q_k^{(3)} &= -\sqrt{5}(2I_k + 1)^{-1}(-1)^{2I_k} \\ &\quad \times \sum_{lm} \left\{ \begin{matrix} 2 & 2 & 2 \\ I_k & I_l & I_m \end{matrix} \right\} M_{kl} M_{lm} M_{mk} \end{aligned} \quad (53)$$

will be needed. The $n = 4$ and 6 moments can be used to give a measure of the β and γ fluctuation of the nuclear shape with the following quantities:

$$\sigma_k(\beta) = \left[q_k^{(4)} - (q_k^{(2)})^2 \right]^{1/2} \quad (54)$$

and

$$\sigma_k(\gamma) = \left[q_k^{(6)} - (q_k^{(3)})^2 \right]^{1/2} \quad (55)$$

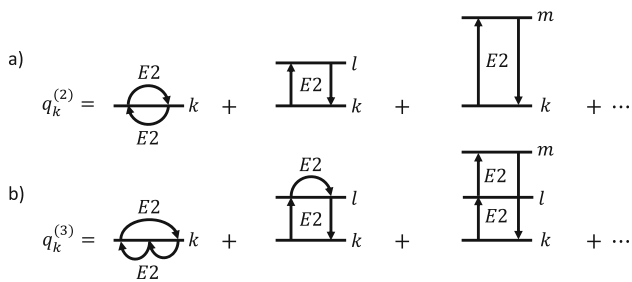


Fig. 3 Schematic illustrations of the developments in terms of reduced electric quadrupole (E2) matrix elements corresponding to the definitions of Kumar's two-body (a - Eq. 52) and three-body (b - Eq. 53) moments. Straight arrows represent off-diagonal elements of the M_{kl} ($k \neq l$) type, while curved arrows indicate diagonal elements M_{kk}

Kumar explains [49] for instance that the ratio $\sigma_k(\beta)/q_k^{(2)}$ would be less than one for a rigid, well-deformed nucleus and larger than one for a nucleus slightly deformable and spherical most of the time.

The 2 and 3-body moments may be discussed in terms of nuclear shape only by assuming an equivalent ellipsoid whose charge, volume, $q_k^{(2)}$ and $q_k^{(3)}$ are those of the nucleus. This is achieved by replacing the definition of the Q_2 operators (Eq. 50) of the n -body moment $Q^{(n)}$ in Eq. 49 by:

$$Q_{2\mu}(k) = (16\pi/5)^{1/2} \int \rho_k r^2 Y_{2\mu} d\tau' \quad (56)$$

This is the μ th component of the quadrupole moment of the equivalent ellipsoid (evaluated in the body-fixed frame) and ρ_k is the charge density equivalent to the one of the nucleus in its state k . One then has:

$$Q_0(k) = \left[(16\pi/5) q_k^{(2)} \right]^{1/2} \quad (57)$$

$$\cos 3\gamma_k = -\sqrt{7/2} q_k^{(3)} \left(q_k^{(2)} \right)^{-3/2}$$

and Kumar shows a relationship with Bohr's β and γ deformation parameters of Eq. 7 in the limit of small deformations [49]. What is interesting with Eq. 57 is that it relates intrinsic shape quantities and the n -body moments $q^{(2)}$ and $q^{(3)}$ which can be evaluated from diagonal and non-diagonal quadrupole electric matrix elements measured in the laboratory reference frame.

These developments were foundational in interpreting Coulomb excitation measurements in terms of nuclear deformation, with the formalism later expanded by Cline [50]. As discussed in the next section, Coulomb excitation experiments played a role in shaping the concept of nuclear structure as pivotal as atomic spectroscopy of hyperfine structures. Interestingly enough, these findings were met with a relative skepticism, particularly among proponents of the rotational model. For instance, Mackintosh notes [8]: "While these ideas are plainly of interest and value, we feel nevertheless

that there are many instances where a specific model (the rotational model) provides a scheme whereby much more information than a single number per multipolarity can be meaningfully used to describe static nuclear shapes. Moreover, $E2$ moments cannot reveal information concerning the neutron distributions."

It is also possible that these doubts arose from the significant anticipated difficulty in measuring all—or at least a sufficient number of—the quantities involved in Kumar's n -body moments. These include the $E2$ strengths of electromagnetic transitions, which must be distinguished from transitions of other electromagnetic nature, as well as the determination of partial widths and quadrupole spectroscopic moments (not only for the ground state but also for excited states), and this for a potentially large number of levels. This is particularly challenging for nuclei presenting complex spectroscopy or high level density (as in odd-mass nuclei). It should also be noted that the reduced transition probabilities $B(E2)$, which are extracted from measurements, correspond to the squared M_{kl} matrix elements required in Kumar's formulae. As a result, their phase cannot be directly determined, which is not an issue for the 2-body moment (as seen in Eq. 52) but becomes problematic for higher-order moments (see Eq. 53). Despite these challenges, and thanks to the continuous improvement of analytical tools [50–52], the community involved in these measurements has nevertheless succeeded in obtaining Kumar's n -body moments beyond $n = 2$ in a few specific cases [52].

2.4 Nuclear translucency

So, by the end of this Section we are seriously touching upon the total impossibility of directly measuring nuclear deformation, even if it were characterizable by a minimal order, quadrupolar, and axially symmetric surface. The profound reason lies obviously in a fundamental principle in quantum mechanics (the uncertainty principle): defining an intrinsic frame and fixing its direction in space leads to an indeterminacy in I , the total angular momentum. Since I is a good quantum number characterizing nuclear states, fixing I means not knowing in which direction the intrinsic frame is oriented. Any deformation mode $\lambda \neq 0$ in Eq. 4 corresponds to angular momentum non-conservation and for odd λ values to parity non-conservation in addition.

As demonstrated by Kumar's method, discussed earlier, the quadrupole moment of a nucleus in its ground state can only be observed through transitions to at least one $I = 2$ state, whose excitation energy is associated with the rotational frequency of a rigid body. Higher-order deformations, such as octupole modes, are also well-documented [24, 53]) and actually crucial for understanding phenomena like fission. The "simplest" (so to say) octupole mode, $\lambda = 3$, $\mu = 0$, corresponds to a static axial octupole deformation—a pear-

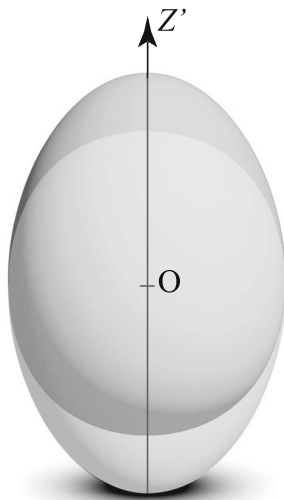


Fig. 4 Schematic view of a nucleus with a static octupole shape: the ground state of such a nucleus appears in the laboratory frame as an average of two orientations of the pear along the symmetry axis (Oz'). In this “exotic” case, the shape of the surface resembles that of an axial quadrupole distribution (although not exactly identical). However, the density distribution is evidently very different, with lower density at the two tips. In the case where the total angular momentum of the nucleus is zero, the intrinsic axis (Oz') is randomly oriented in space, and the shape appears essentially spherical in the laboratory frame, as shown later in Fig. 6 (nuclear translucency)

shaped nucleus. Analogous to quadrupole deformations, the ground state of such a nucleus appears in the laboratory frame as an average of the two orientations of the pear along the symmetry axis (Oz') (this is illustrated highly schematically on Fig. 4). Intrinsic odd electric moments like $E1$, $E3$, etc., manifest only through transitions to opposite-parity states, located at excitation energies associated with the tunneling frequency between two orientations of the pear. These tunneling paths involve higher-energy configurations that preserve axial symmetry. In the intrinsic frame, the octupole state is a single mixed-parity state, whereas, in the laboratory frame, it appears as a parity doublet of states with the same spin and opposite parity. Ideally, these states are degenerate in energy, although, in practice, various mechanisms often lead to a significant lifting of this degeneracy [24, 53].

And so the “quantum fairy” casts her curse upon the concept of nuclear shape from its very inception—the even-even nuclei all having a total angular momentum of zero in their ground state. This true “quantum fog” also has another unfortunate consequence, which we will illustrate now: it mixes radial and angular distributions as observed in the laboratory frame. Consider, for instance, a uniform distribution bounded by a sharp nuclear surface (no radial dependence) like the revolution ellipsoid for an even-even nucleus in its ground $I = 0$ state. What is observed from the laboratory frame is an average over all possible orientations of the intrinsic

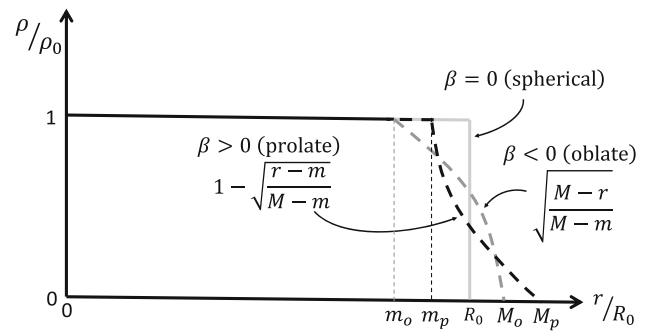


Fig. 5 Uniform distributions bounded by a sharp nuclear surface of axially symmetric quadrupolar ($\beta > 0$ prolate or $\beta < 0$ oblate) shape averaged over all possible orientations of the intrinsic frame

frame. Averaging over all angles, the resulting density takes the form [54]:

$$\rho(r) = \rho_0 \times \begin{cases} 1 - \sqrt{\frac{r - m_p}{M_p - m_p}} & \beta > 0 \\ \sqrt{\frac{M_o - r}{M_o - m_o}} & \beta < 0 \end{cases} \quad (58)$$

where m_o (m_p) defines the radius up to which $\rho = \rho_0$ for an oblate (prolate) deformation, and M_o (M_p) the maximal extent of the nuclear density for an oblate (prolate) deformation:

$$\begin{aligned} m_p &= R_0 \left(1 - \frac{1}{2} \sqrt{\frac{5}{4\pi}} \beta \right) \\ m_o &= R_0 \left(1 - \sqrt{\frac{5}{4\pi}} |\beta| \right) \end{aligned} \quad (59)$$

and

$$\begin{aligned} M_p &= R_0 \left(1 + \sqrt{\frac{5}{4\pi}} \beta \right) \\ M_o &= R_0 \left(1 + \frac{1}{2} \sqrt{\frac{5}{4\pi}} |\beta| \right) \end{aligned} \quad (60)$$

These apparent density distributions are represented in Fig. 5. It is very clear that averaging the uniform distribution over all angles leads to the emergence of a radial dependence in a region extending from $m_{o/p}$ to $M_{o/p}$ at the surface of the nucleus. This phenomenon is referred to as “nuclear translucency” in the literature. The extent of this translucent border $M_{o/p} - m_{o/p}$ is directly proportional to the quadrupole deformation parameter β . This effect adds to the true nuclear radial distribution and makes it particularly difficult to determine for highly deformed nuclei. It is also evident how nuclear

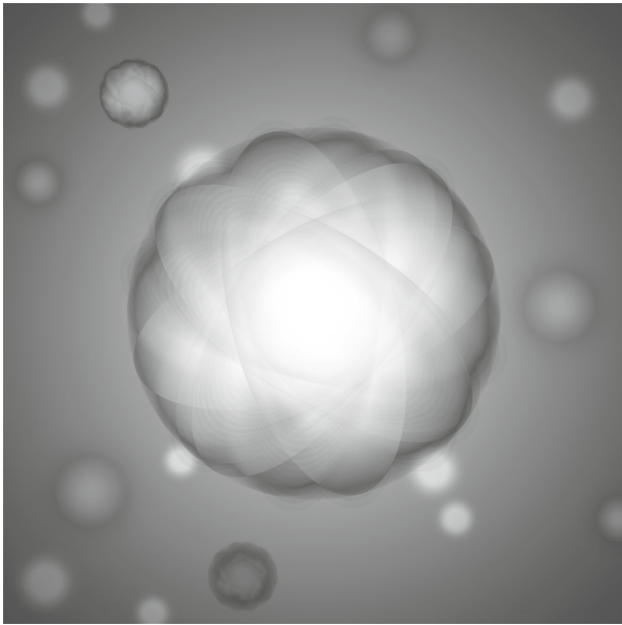


Fig. 6 Illustration of the nuclear translucency phenomenon: a target consisting of even-even nuclei in their ground state with $I = 0$, exhibiting a pronounced prolate shape ($\beta \approx 0.3$), as it would appear to an artist seated on an incident particle or an electron from the surrounding atomic cloud (the shape appears to them as averaged in space over all possible orientations of the symmetry axis, resulting in the appearance of a sphere with a blurred outline)

deformation confers a charge extension to a nucleus, which will have a noticeable effect on the isotopic shifts of the hyperfine spectrum centroids, one of the very first experimental indicators of the existence of nuclear shape as we will see in the next section.

Finally, in Fig. 6, an artist's rendition is provided of what a strongly deformed prolate target nucleus in its ground state $I = 0$ should look like when viewed in the laboratory frame: a sphere, with a slightly blurred edge (the nuclear translucency).

3 Genesis and history of the nuclear shape concept: key milestones

In this Section, we briefly revisit the key steps that marked the genesis and traced the history of the nuclear shape concept. This history is marked by the emergence of three highly influential reviews on the subject, Refs. [4, 8, 55], to which the reader is strongly encouraged to refer, along with the abundant literature cited therein. For the present work, the author deliberately sets aside the details of theoretical developments (addressed elsewhere in this series of articles), focusing instead on aspects most originally linked to the evolution of the concept of nuclear shape or related to the identification of mechanisms that could lead to the emergence of

nuclear deformation. On the experimental front, the literature on nuclear shapes is vast, and the selection of works presented here aims to introduce the major classes of experiments and observables that have contributed to the emergence of the nuclear shape concept or illustrate its ability to correlate a wide range of measurements of very different natures. The general idea is to provide a correct perspective on the historical event constituted by the advent of ultra-relativistic heavy-ion collisions in this field.

3.1 1st Period 1909–1950: from Rutherford to the Rainwater–Bohr revolution

3.1.1 1909–1936 - From α scattering to Eu/Sm isotope shift

Any conception of nuclear shape inevitably requires the realization that the nucleus is not a point-like object, but rather possesses a finite size. This realization was made possible by the alpha particle scattering experiments conducted by Geiger and Marsden [56, 57], aptly interpreted by Rutherford [58], and expounded upon in the monograph “Radiation from Radioactive Substances” [59]. From that moment, when there was also a growing ability to measure atomic mass with increasing precision, and thus total binding energy of atomic nuclei, arose the formidable question of the mutual relationship between these two quantities — what came to be known as “nuclear force saturation properties.” From the simple observation that both nuclear volume and total binding energy increased linearly with the number of nucleons A (rather than with the number of possible nucleon pairs $A(A+1)$ as would have been expected for the binding energy) emerged the first conjectures about the nature of this mysterious interaction ensuring the cohesion of a system of A nucleons. However, none of these considerations necessitated the notion of nuclear shape, and there was nothing in the properties of the nuclear interaction (as they could be imagined at the time) that would push the system to deform. Surprisingly, this remained the case even when, a little later, Rabi's team at Columbia University in 1939 demonstrated the existence of a quadrupole moment for the deuteron [60, 61], though this discovery actually paved the way for highlighting the importance of tensor terms in the nuclear force (especially in the proton-neutron channel) and their ability to drive the nucleus toward deformation.

Even though molecular spectroscopy had advanced enough at the time to envision that a molecule like ethane could be a non-spherical quantum object capable of rotation and vibration, the energy resolutions available in the 1930s in experiments allowing nucleus excitation were not sufficient to separate excited states. Consequently, there was no indication that the first excited states of an even–even nucleus could be described as the first states of rotation of an intrinsic shape. It was from the field of physics with the highest degree of

precision in its measurements, atomic spectroscopy, that the first results indicating that the atomic nucleus could possess a non-zero electric quadrupole moment came. The saga of quadrupole moments arising from hyperfine structure measurements is very exhaustively described in the review of Peter Brix [62] who was one of its protagonists, and one can also find a wealth of information in Lieb's well-documented paper [63]. Schüler and Schmidt, studying the hyperfine spectra of rare earth samples at the Potsdam Observatory, discovered significant deviations from the Landé interval rule in the isotopes $^{151,153}\text{Eu}$. These deviations could not be attributed to perturbations of atomic levels, as sometimes observed when the hyperfine splitting is close to the fine level spacing. As P. Brix cunningly recalled [63], citing one of his mentors: "fortune favours the prepared mind" and Schüler and Schmidt must have been aware of 1931 Racah's work [64] showing that a nucleus elongated by one tenth (1 fm) of its radius would cause deviations from the Landé interval rule of the order of the magnetic hyperfine splitting in $^2P_{3/2}$ levels (an unpublished suggestion by Pauli to this effect is also mentioned at various points in the literature). So they interpreted their discovery as a new phenomenon in a now famous article [65] titled "Über Abweichungen des Atomkerns von der Kugelsymmetrie" ("On deviations of the atomic nucleus from spherical symmetry"), which is considered the birth of the concept of nuclear deformation. They observed the same effect in ^{175}Lu and in the related publication [66] first appeared the term "electric quadrupole". Schmidt explains [67] "The simplest assumption which could explain the observed perturbation was a nucleus elongated or compressed along the spin axis. During my studies I had never attended a lecture on electrodynamics and it was only Mr. Dellbrück in Berlin who explained to us what a quadrupole moment is". Casimir upon a visit to Schüler in Potsdam immediately provided the theory in Ref. [27] which contains the first expression of the hyperfine structure term (the one perturbing the Landé interval rule) originating from the coupling to Q_s . The year after, in 1936, in his prize essay for Teyler's Foundation at Haarlem [28], Casimir already foresaw the entire course of history with astounding scientific clairvoyance: "then one is led to the conclusion that the quadrupole moment cannot possibly be due to one proton. So it is necessary to assume that it is caused by a group of particles. It might even be that the nucleus as a whole has a prolate shape and that the nucleus as a whole is rotating about its major axis. In this connection there may be some point in remarking that for heavy nuclei the rotational energy estimated by means of the elementary formula $E = \frac{\hbar^2}{2\mathfrak{I}} I(I+1)$ ($\mathfrak{I}$ = moment of inertia) is small compared with the binding energy. In any case the existence of large quadrupole moments seems to offer an argument in favour of the idea that the nucleus must be treated as a whole and that its prop-

erties are not determined by the behaviour of one particle". In their samples of rare earths, Schüler and Schmidt also had samarium. They measured the isotopic shift between the different isotopes and found a larger change between ^{150}Sm and ^{152}Sm . Now, these two nuclei are isotones of ^{151}Eu and ^{153}Eu , which have just been discussed at length, and, as addressed in Sect. 2, nuclear elongation along an axis should have a significant effect on the isotopic shift. However, this connection was not made because it presupposed the notion of an *intrinsic* quadrupole moment (the Q_0 of Sect. 2). Generally, outside the small circle of pioneers just mentioned, the idea that the nucleus could be something other than spherical was far from generally accepted. The comments made by Bethe and Bacher [10] on these discoveries in their 1936 review are eloquent in this regard: "Schüler and Schmidt observed that the deviations could be satisfactorily accounted for by the presence of an interaction term which is proportional to the square of the cosine of the angle between the nuclear spin and the extranuclear angular momentum. Such an interaction, they pointed out, would be expected if the nucleus has an electric quadrupole moment. It is indeed reasonable that nuclei should have *small* electric quadrupole moments. Such a moment arises if the protons are not distributed over the nucleus on the average with spherical symmetry".

In fact, it was not until the post-war work of Brix and Kopfermann in Göttingen [68] that this irregularity in the sequence of isotopic shifts of Sm isotopes was linked to the existence of an intrinsic quadrupole moment for the ground states of even-even nuclei with a total angular momentum of zero (despite this fact!). Their revolutionary idea was to assume that the $I = 0$ ^{150}Sm and ^{152}Sm had about the same intrinsic deformation as ^{151}Eu and ^{153}Eu , and, using the values of the Eu quadrupole moment, calculate how much this increase in deformation contributes to the (volume effect) isotope shift. And the result was correct.

We will conclude this part of the history of the concept of nuclear shape by noting that despite these extraordinary discoveries, the emergence of this concept undoubtedly suffered from an epistemological bias that significantly delayed its acceptance within the scientific community of the time. This epistemological bias had been introduced by the fact that large quadrupole moments had been observed first and most frequently in odd-proton nuclei, and were thus associated with a proton-specific effect, probably induced by the unpaired proton. Furthermore, there was nothing yet known about the internal structure of nuclei that would suggest that, even if they could potentially deform, this deformation was static. The reason, which we will develop in the following paragraph, is essentially that the well-studied saturation properties at the time suggested that a natural model for the nucleus was that of the liquid drop, in which the only deforming effects came from the Coulomb repulsion between protons. This epistemological bias had additional, deeper origin

rooted in the principles of quantum mechanics themselves. Let's hear Brix himself on this matter [62]: “The idea of intrinsic quadrupole moments for $I = 0$ nuclei was at first not generally accepted. It was said that they cannot be measured and therefore do not exist. I remember that C. F. von Weizsäcker at that time made the encouraging remark that ‘an elephant in a s -state is spherically symmetric but still remains an elephant’. When I met von Weizsäcker recently, he recalled at once that he had said this, but added that the example came from Heisenberg”.

3.1.2 1936–1950: from the liquid drop model to the Rainwater-Bohr revolution

In parallel with the developments just mentioned, Fermi and collaborators in 1934 irradiated nearly all known stable isotopes with neutrons [69]—including uranium, which turned out to be a special case we will revisit shortly. The neutron capture cross sections and neutron emissions following the capture were brilliantly interpreted by Niels Bohr within the framework of his compound nucleus model [70], which received a triumphant reception in the years 1936–37. The observation of a large number of very narrow resonances (< 1 eV) closely spaced in energy (tens to hundreds of eV) in the interaction cross-sections between low-energy neutrons and intermediate to heavy nuclei from these early historical measurements inspired N. Bohr to draw an analogy with the absorption of a molecule at the surface of a liquid droplet. In this picture, the long lifetime of the “compound” nucleus state, consisting of the target nucleus and the projectile neutron, is interpreted as resulting from the energy initially concentrated in the incident particle being distributed among all the constituents of the system. “Each of the particles of the *compound nucleus* will have some energy, but none will have sufficient energy to escape from the rest. Only after a comparatively long time, the energy may *by accident* again be concentrated on one particle so that this particle can escape” explains Bethe [71]. Concurrently, the liquid drop model introduced by von Weizsäcker in 1935 [72] to calculate binding energies also met with great success. In Berlin, Hahn, Meitner, and Straßmann, while attempting to understand their observations during the irradiation of uranium with neutrons, finally in 1938 discovered traces of radioactive barium in the reaction residues. Meitner and Frisch [73], with the nucleus conceived as a charged liquid drop in mind, proposed to explain the origin of this phenomenon by suggesting that neutron capture causes such a large oscillation of the nucleus surface that it eventually splits into two pieces, as illustrated in Fig. 7—and Frisch coined the term “fission” for this phenomenon. It was therefore natural that, following this discovery, Bohr and Wheeler proposed a model [74] for the fission mechanism as an extension of the compound nucleus model, firmly anchoring this phenomenon in the conception

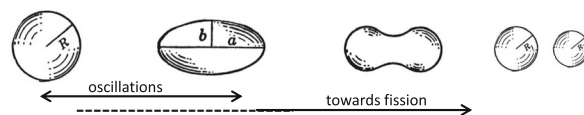


Fig. 7 Illustration of the fission process, viewed as resulting from oscillatory modes of a charged liquid drop (Bohr and Wheeler model), leading transiently to extreme nuclear deformations

of the nucleus as a charged quantum liquid drop homogeneously charged with Z positive charges.

Within the framework of the liquid drop model, nothing necessitates (let alone explains) that the ground state of a nucleus can exhibit a static shape: the only repulsive component of the model is the Coulomb force between protons, which, if too many, eventually leads to a limit of existence against fission governed by a certain critical parameter proportional to Z^2/A . But there is no room for deformation in the ground state other than transiently, via oscillation with the surface tension of the liquid drop as the restoring force: to undergo fission, one must pass through strongly elongated shapes as seen in Fig. 7, but these oscillatory modes are associated with excited states.

And, in an amusing twist of history, the missing ingredient that would allow the crystallization of the concept of (static) nuclear (ground state) deformation and its full utilization in models came from the (spherical!) shell model. Indeed, the discovery of the neutron in 1932 [75] immediately shed new light on the internal structure of the nucleus—a structure that is denied by the liquid drop model, as we have just seen, the compound nucleus is a structureless system. In the same year, Bartlett concluded [76] that “whether or not the α -particle plays a fundamental part remains to be seen, but we can certainly disregard it for the lightest elements (!) and consider protons and neutrons as being the elementary constituents”, and suggested a structure by “analogy with the external electronic system”, already referring to ^{16}O as a “formation of some sort of closed shell.” The discovery of the neutron clearly laid the foundation for what would later be called a “microscopic approach” (by opposition to the liquid-drop-model based approaches referred to as “macroscopic”), which involves reducing nuclear properties to individual degrees of freedom, chosen early on to be those of nucleons, or more precisely “individual particles” evolving in an average potential. The beginnings were challenging: Guggenheimer [77] and Elsasser [78,79] for instance sought the appropriate forms of potential to explain the “special numbers” of nucleons (which would become the “magic numbers”, a term coined later on by Wigner to show his scepticism [80]) as due to shell closures. However, this approach largely fell to the wayside in the latter half of the 1930s. This shift was due to several factors: (i) the difficulty in reproducing the observed magic numbers without overly artificial

of the total angular momentum I , the maximum observable quadrupole moment is reduced by the factor $\mathcal{P}_I$ from Eq. 40 above. Rainwater continues, saying that “After Professor Townes’s colloquium, Dr. Bohr and I had many discussions about my concept. He was particularly interested in the dynamical aspects. The distortion bulge could, in principle, vibrate or move around to give the effect of rotational levels. The first result was his January 1951 paper [42], ‘On the Quantization of Angular Momenta in Heavy Nuclei’.”

During this brilliant encounter, the two major axes of the rich phenomenology associated with nuclear shape were already being drawn. The first is, of course, associated with the idea of an intrinsic quadrupole moment, which is naturally not observable for the ground state $I = 0$ of even-even nuclei but should reveal itself when the nucleus leaves its ground state for the first excited 2^+ level via a quadrupole excitation. This avenue of investigating nuclear shape was quickly explored because early accelerators allowed for Coulomb excitation experiments [92] (the associated concepts have been discussed in Sect. 2.3).

The second major axis concerns the origin of nuclear angular momentum. Indeed, the wave function of the unpaired particle, creating the “bulge” mentioned by Rainwater in a deformation-driving mechanism, will have (for example, for an elongated prolate spheroidal potential along the z' -axis) a dependence in φ' (the azimuthal angle in the intrinsic reference frame) for $l_{z'} = m$ as $e^{im\varphi'}$. However, $l_{x'}$ and $l_{y'}$ and $\vec{l}^2$ cannot be good quantum numbers; the core must somehow share the net angular momentum. The exploration of this other research axis—how the spin of nuclei is generated, and up to what angular momentum, seen as corresponding to core rotation, the nucleus can withstand—would also become possible thanks to advances in heavy-ion accelerators, which allowed for fusion-evaporation experiments resulting in residual nuclei in states of very high angular momentum [93]. These two axes would be at the heart of the subsequent period, reaching their climax with the discovery of superdeformation.

But before opening this new chapter, it is curious to note that Rainwater’s adventure with nuclear shapes stops there, and he would only participate indirectly through his (pioneering [94]) research on X-ray emissions from muonic atoms (which, incidentally, have more to do with nuclear radius than with the nuclear shape itself).

It is also important to cite one of Rainwater’s final sentences in his Nobel acceptance speech: “Although my consideration of the ‘forcing term’ for spheroidal nuclear distortion considered the dependence of the single-particle kinetic energy on the distortion, I have never seen a description of my work elsewhere in those terms. A common equivalent phrasing is the ‘centrifugal force exerted on the barrier’ by the orbit.” Indeed, such vivid physical imagery was in the minds (and abundantly used: “particle pressure on the sur-

face,” centrifugal forces, Coriolis forces, perturbation of particle motion by “capillary modes” of the surface motion, etc.) of those who developed what would later be called the collective model or “unified model,” particularly the Copenhagen school. This is a notable and probably unique fact about that generation of nuclear physicists: these images guided their theoretical approaches (a quick glance at the extraordinary figures of Ref. [90], like those reproduced in Fig. 9 here, is enough to be convinced of this). “Give me a vivid picture of the behavior of the wave function,” to quote Wigner from the preamble of this article. Obviously, this picturesque mode of operation is no longer in vogue and probably would not be tolerated in today’s literature.

Finally, on a conceptual level, it should be noted that by the end of this period, deformation and shell structure had become inseparable. Nuclear deformation *requires* the existence of a shell structure. Even for the most extreme deformations, as it was quickly realized that it was impossible to accurately calculate a fission barrier using the liquid drop model without including shell effects. This would be magnificently formalized a bit later by Myers and Swiatecki [95].

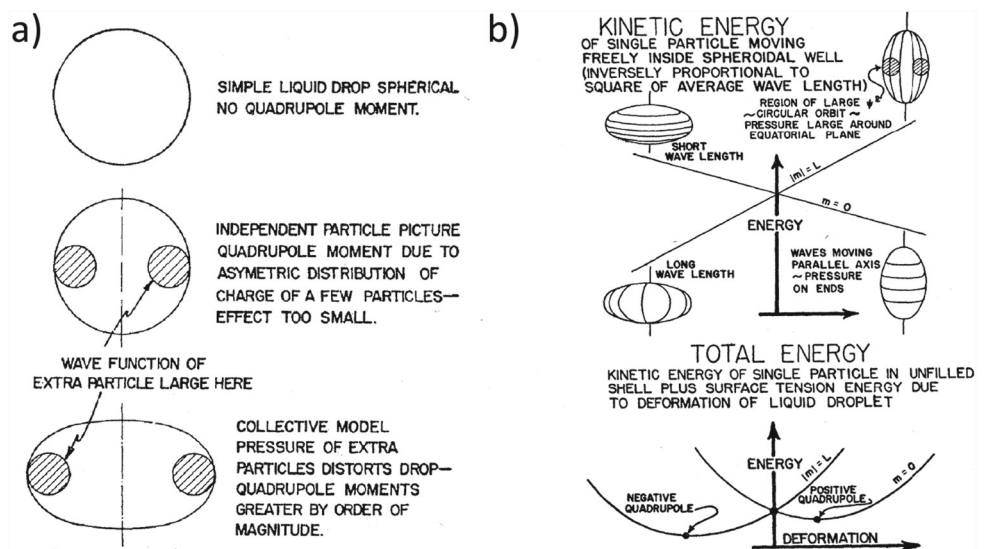
3.2 2nd period 1951–1985: from the unified model to superdeformation

In the wake of Rainwater’s proposal for a spheroidal nuclear model, A. Bohr [26], along with and followed by many others, notably B. R. Mottelson [18] and S. G. Nilsson [96, 97], founded the collective model. This model is also known as the unified model because it unified the description of individual and collective degrees of freedom (in other words, as we have seen previously, the best of both worlds: the shell model and the liquid drop model). The basic hypothesis of this model is that the nucleus can be described as a shell structure capable of oscillations in shape and size. The system exhibits many analogies to molecular structures with the interplay between electronic and nuclear motions. Exploiting this analogy, Bohr and Mottelson [18] proposed relating the individual and collective degrees of freedom in the simplest way, by describing the nucleus with the wave function:

$$\Psi_{\Omega R}(\xi, x) = \phi_R(\xi) \cdot \chi_{\Omega}(x) \quad (61)$$

where x represents the individual (space, spin, isospin) coordinates of all particles in the nucleus $\chi_{\Omega}(x)$ is the “intrinsic” (shell model) wave function specified by a set of quantum numbers Ω for a fixed (deformed) potential characterised by the parameters ξ and $\phi_R(\xi)$ describes the motion of the nucleus as a whole, characterized by additional quantum numbers R . This description would be accurate only in the so called “adiabatic” limit, which correspond to the case where the frequencies ω_p for particle excitation are large compared with the frequencies ω_c of the collective type of

Fig. 9 **a** Figure 5 and **b** Fig. 6 of Ref. [90] reprinted with permission from D. L. Hill and J. A. Wheeler, Physical Review, Vol. 89, No. 5, p. 1102 (1953) Copyright 2025 by the American Physical Society (License Number: RNP/25/JAN/087743). These figures illustrate the mechanism of nuclear surface distortion caused by the motion of an individual particle, as conceived by Rainwater on kinetic energy arguments (see text), leading to a permanent deformation of the entire nucleus



motion [18], or said in other words, that the single-particle excitation energies are large compared with the energy of the lowest collective excitation (e.g. the energy of the first 2^+ state in even-even nuclei), or in case of collective rotational motion, that the rotational velocity of the nuclear surface is slow compared with the nucleon velocity inside the nucleus. Hill and Wheeler [90] verified this hypothesis for the case of a nucleon with a kinetic energy of 15 MeV moving within a ^{236}U nucleus, and we reproduce their calculation here for convenience. The characteristic time of radial motion is obtained by evaluating the following integral [90]:

$$\begin{aligned}
 t_{\text{nucleon}} &\approx \int_0^R \left[\frac{2T}{M} - \frac{l(l+1)\hbar^2}{M^2 r^2} \right]^{-1/2} dr \\
 &= \frac{2R}{(2T/M)^{1/2}} \left[1 - \frac{l(l+1)\hbar^2}{2MT R^2} \right]^{1/2} \\
 &< \frac{2R}{(2T/M)^{1/2}} \equiv \frac{2R}{v} = \frac{2r_0 A^{1/3}}{0.18c} \\
 &= 0.3 \times 10^{-21} \text{ s}
 \end{aligned} \quad (62)$$

where M is the mass of the nucleon, T the kinetic energy term, l its orbital momentum quantum number, R the nuclear radius taken as $r_0 A^{1/3}$. This is an order of magnitude smaller than the body rotation time associated to the first 2^+ excitation state [90]:

$$\begin{aligned}
 t_{2^+} &= \frac{2\pi\hbar}{\hbar\omega_2} \\
 &\approx \frac{2\pi \times 0.658 \times 10^{-21} [\text{MeV} \cdot \text{s}]}{0.8 [\text{MeV}]} \\
 &= 5 \times 10^{-21} \text{ s}
 \end{aligned} \quad (63)$$

We are far from the particle that creates a bulge in Rainwater's deformation-driving mechanism, as this mech-

anism involved the particle's motion exerting pressure on the nuclear surface—a highly non-adiabatic effect. This is an interesting paradox to note in the history of the nuclear shape concept. Rainwater was correct in stating that Bohr was more interested in the dynamic aspects of the problem, considering the rotational and vibrational movements associated with the bulge. Once the hesitation regarding the existence of an intrinsic nuclear shape was overcome, the question of the origin of this deformation took a back seat in favor of its practical applications. This shift allowed for the (very efficient) absorption of many individual degrees of freedom into the collective degrees of freedom that could be associated with the shape motions. It was not until the advent of self-consistent calculations that the question of why a particular nucleus adopts a specific shape was satisfactorily addressed once again. In this respect it is also interesting to note that during the concerned period, the supreme proof of the existence of an intrinsic nuclear shape was considered to be the good agreement between quantities calculated by the collective model and experimental results. Nuclei that did not exhibit the properties predicted by this model were considered shapeless, rendering the concept of shape irrelevant in their context. For instance, Mackintosh in his 1977 review [8] says “static deformations are never less than $\beta \approx 0.2$ [...] nuclei which have neither $\beta = 0.0$ nor $\beta \approx 0.25$ static deformations ($\beta \approx 0.4$ for $A < 40$) are *soft*” and cannot be described by shell-model type intrinsic states (the χ_Ω 's of Eq. 61).

If we consider the case where the collective degrees of freedom are associated solely with the rotation of an axial nuclear surface of the type shown in Fig. 2—the famous axial rotor—and where the mean potential follows the matter distribution in the intrinsic reference frame, whose eigenstates are the wave functions χ_Ω , the total wave function of the

nucleus Eq. 61 must be correctly symmetrized and takes the following “canonical” form [41]:

$$\begin{aligned} \psi_{IM}^K(\theta, \varphi, x) = & \sqrt{\frac{2I+1}{16\pi^2}} \\ & \times \left[\mathcal{D}_{MK}^I(\theta, \varphi) \chi_\Omega(x) + (-1)^{I+K} \mathcal{D}_{M-K}^I(\theta, \varphi) \chi_{-\Omega}(x) \right] \end{aligned} \quad (64)$$

where $\mathcal{D}_{MK}^I(\theta, \varphi)$ are the rotation Wigner D-matrices, giving the amplitude for the intrinsic state χ_Ω to be oriented at the Euler angles (θ, φ) , and the second term in the brackets is there to make the wavefunction invariant to an inversion in the choice of the intrinsic ($0z'$) axis. The wave functions like those of Eqs. 61 or 64 are indeed vivid wave functions, as Wigner liked, and foundational for the very fruitful exploitation of the concept of nuclear shape. However, they introduce a formal difficulty that was discussed almost immediately after their introduction. To quote Peierls and Yoccoz [98]: “if one tries to use simply the surface variables and the particle coordinates as dynamical variables, one has a redundant set; a formalism using redundant variables is possible, but liable to lead to mathematical complications”. The problem of the redundant coordinates has given rise to a large amount of work and literature at the time (see e.g. Ref. [99]). But even with this flaw in the background, the collective model proved to be very effective and highly inspiring.

One of the essential implications of the collective model from a spectroscopic perspective corresponds to the realization of Casimir’s earlier vision: the appearance of bands of rotational states, with energy sequence in $E_I = \frac{\hbar^2}{2\mathcal{I}} I(I+1)$. Each band corresponds to different rotational states of an intrinsic state (characterized by a shape in the intrinsic frame and a given Q_0) that persists across different I states. Intra-band transitions are then very strong, and the transition probabilities are all related to the unique Q_0 by geometric factors like Clebsch-Gordan coefficients. The intrinsic state, in a sense, inherently contains the entire rotational band. Because, as noted earlier, fixing the orientation of the intrinsic state implies that the total angular momentum I is not determinable, so the intrinsic state is more a wavepacket of states with different angular momentum values, a superposition of all the states of the rotational band “built on top of it”. One may thus project out of it states with total angular momentum I and projection quantum number M on the laboratory-frame $0z$ axis, and obtain the wave function [8, 98]:

$$|\psi_{IM}^K\rangle = \frac{2I+1}{8\pi^2 N_I^{1/2}} \int d\omega \mathcal{D}_{MK}^I(\omega) \hat{R}(\omega) |\chi_\Omega(x)\rangle \quad (65)$$

where $\hat{R}(\omega)$ rotates the intrinsic state $|\chi_\Omega(x)\rangle$

$$\hat{R}(\omega) = \exp(-i\alpha I_z) \exp(-i\beta I_y) \exp(-i\gamma I_z)$$

and

$$N_I = \frac{2I+1}{8\pi^2} \times \int d\omega \mathcal{D}_{KK'}^I(\omega) \langle \chi_\Omega | \hat{R}(\omega) | \chi_\Omega \rangle$$

is the normalization.

The authors of that period considered the extent to which the matrix elements of various operators (typically electric transition operators) acting between these projected states corresponded to those obtained using wave functions of the type given by Eq. 64, derived from the same intrinsic state χ_Ω , as the measure of the validity of the adiabatic rotor model. If the correspondence is good, then the density of the intrinsic wave function χ_Ω can be considered as characterizing the intrinsic shape of the nucleus.

It is beyond the author’s capacity to determine if the (picturesque) condition of adiabaticity formally translates to the condition of proximity between the values of matrix elements evaluated with the wave functions of Eqs. 64 and 65, but it turns out that the two arguments roughly delineated the same zones of validity: the collective model, and thus with it the concept of shape, were reputed during those years to be better defined for nuclei in the rare earth region, the actinides region, and beyond. A light nucleus like ^{20}Ne , although presenting a spectroscopy with a $I(I+1)$ dependence, was considered a nucleus to which the concept of shape could not be applied, as this dependence was considered to arise from another mechanism (in this case, α clustering). This certainly played a role in shaping experimental research programs.

S.G. Nilsson proposed as early as 1955 intrinsic wave functions χ_Ω (of Eq. 64), which are eigenfunctions of a deformed axial mean potential [96]. However, at that time, the spherical shell model was also gaining traction, raising the question of the equivalence between the two approaches. From an epistemological (reductionist) standpoint, the spherical shell model is more legitimate than the unified model in the sense that it is more microscopic. The calculations explicitly take into account the interaction between nucleons within a valence space and are performed in the laboratory frame, where observables such as the spectroscopic quadrupole moment are computed and can be directly compared with experimental measurements—without invoking the concept of nuclear shape and the symmetry breaking that it entails.

Nevertheless, given the remarkable success of the rotational model, the question of the equivalence between the two approaches naturally arose, and J.P. Elliott [100] provided an elegant answer by employing symmetry principles within the spherical shell model. Considering the success of the rotational model applied to ^{19}F (which could also be treated using early shell model calculations), Elliott wrote [100]:

“The situation is therefore a challenging one. It suggests that there is some connexion between the shell model as applied to these light nuclei on the one hand, and Nils-son’s version of the rotational model on the other. In particular, it implies that the shell model wave functions for ^{19}F , which are the result of configuration mixing, bear some close relation to rotational wave functions. If this is indeed the case, then there should be some underlying ideal coupling scheme which exhibits the rotational properties ab initio. Just as the concept of seniority brings out the single-particle features for states of particles in a pure configuration, we must now look for some quantum numbers which bring out the rotational features for states of particles in mixed configurations. In particular, we shall be interested in the mixed configurations of degenerate oscillator orbits because, at least for the light nuclei, this is likely to be the most important mixing. A search for possible quantum numbers is necessarily a search for the possible classification schemes, or in other words, a study of the possible transformation properties of the wave functions. The angular momentum L may be looked upon as describing the transformation properties of the wave function under three-dimensional rotations in coordinate space. We must study more general groups of transformation than this. Fortunately, Racah has developed very powerful methods for such a study, and although he has only applied them to pure configurations, they may also be used for mixed configurations.”

With this approach the typical $I(I + 1)$ energy dependence of the spectra is obtained without dropping rotation invariance, and the notion of an intrinsic state appeared associated to the choice of the chain of subgroups of $\text{SU}(3)$ used to label the eigenfunctions of the problem: $\text{O}(3)\text{-U}(1)$ for the states in the laboratory frame and $\text{SU}(2)\text{-U}(1)$ for intrinsic states. In the latter case, a new quantum number emerge which is nothing more than an intrinsic quadrupole moment [101, 102].¹²

¹² Elliott’s legacy would later be extensively pursued by the Strasbourg-Madrid group [102–104], who explored several variants of his model (pseudo- $\text{SU}(3)$, quasi- $\text{SU}(3)$), notably in the context of shape coexistence studies, which we will discuss when addressing the final historical period in Sect. 3.3. Of note, except in a few clear cases where a distinctly rotational structure emerges in observables computed using the spherical shell model—allowing for the application of e.g. Equation 38 to infer a possible intrinsic shape corresponding to these results—the discussion of shell model results in terms of nuclear shapes remained quite limited for a long time. This posed a particular challenge for experimentalists in interpreting their data for nuclei with weakly pronounced or unstable shapes. Consequently, it was the experimentalists themselves [105] who took the initiative to process the electromagnetic matrix elements obtained within the shell model framework (in the laboratory frame) using the Kumar formalism described in Sect. 2.3, in the same manner as they would have for experimental data—albeit with the advantage of directly accessing their relative phases—to infer the shape of the computed states. The information obtained via this method corresponds

As noted earlier, we can only ensure the validity of the scheme depicted in Fig. 2 if the same value of Q_0 consistently produces different measured values in the laboratory frame. The collective model relates (intra-band) transition probabilities to the Q_0 of the intrinsic state (hence β the deformation parameter for instance via Eq. 47) by geometric factors. For instance, the reduced transition probability $B(E2; I_i 0 \rightarrow I_f 0)$ for an electric quadrupole transitions from state I_i to state I_f in the $K = 0$ band of an even-even nucleus is given by [108]:

$$B(E2; I_i 0 \rightarrow I_f 0) = \frac{5}{16\pi} e^2 Q_0^2 \langle I_i 200 | I_f 0 \rangle^2 \quad (66)$$

which reduces, for the 0^+ ground state to first excited 2^+ state transition, to:

$$\begin{aligned} B(E2; 0^+ \rightarrow 2^+) &= \frac{5}{16\pi} e^2 Q_0^2 \\ &= 5 \cdot B(E2; 2^+ \rightarrow 0^+) \end{aligned} \quad (67)$$

And as soon as they became feasible with the advent of the first accelerators, Coulomb excitation cross-section measurements immediately established themselves as a primary tool for studying nuclear shape (Ref. [92] is a early, seminal article in this field) and remain so in contemporary times [52] (now utilizing radioactive ion beam accelerators [51]).

The internal coherence of the collective model, and consequently the “tangibility” of the existence of an intrinsic quadrupole moment, could then be quickly tested e.g. by comparing the deformation parameter β extracted from spectroscopic quadrupole moments obtained through the study of hyperfine spectra, as previously discussed, with those extracted from the intrinsic quadrupole moment Q_0 in Eq. 67, derived from entirely different experimental data. The coherence obtained between the two was quite remarkable for the regions of rare earth elements and transuranium elements. This extraordinary success solidified the collective model as a canonical model, thus embedding the concept of nuclear shape into the routine thinking of nuclear physicists as an evident reality.

Nevertheless, the coefficient e in front of Q_0 in Eq. 67, representing the elementary charge, serves as a reminder that in Coulomb excitation experiments (and this also applies to hyperfine spectrum measurements), it is the charge distribution within the nucleus that is probed. The intrinsic quadrupole moment obtained is an electric moment. However, as previously discussed, the collective model includes

to the shape of the homogeneously charged ellipsoid equivalent to the nucleus, which would reproduce the given set of electromagnetic matrix elements (not yet the exact nuclear shape). This approach was studied more systematically somewhat later by Poves et al. [106]. Finally, it is worth noting that the Tokyo group introduced a “T-plot method” to analyze nuclear shapes in a large scale Monte Carlo Shell Model approach employing wave functions defined in an intrinsic frame [107].

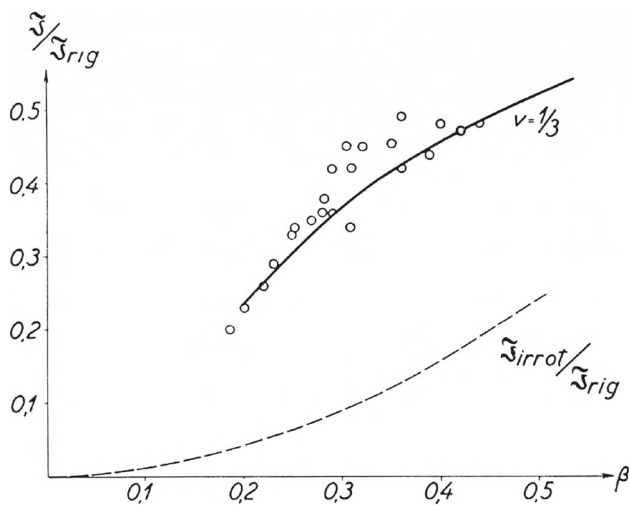


Fig. 10 Figure reprinted from “Moments of Inertia of Rotating Nuclei” in: A. Bohr and B. R. Mottelson, *Matematisk-fysiske Meddelelser*, bind 30, no. 1, 1955 (Det Kongelige Danske Videnskabernes Selskab, pub.royalacademy.dk 2025) [108]. This historical figure (abundantly reproduced in the subsequent literature) shows the dependence of nuclear moments of inertia on the nuclear deformation parameter β . The full-drawn curve represents a theoretical estimate (based on the two-nucleon model developed in Ref. [108], with an interaction parameter $\nu = 1/3$, see therein for further explanation). The moment of inertia corresponding to irrotational flow is shown by the dotted curve

the fact that in the case of rotational motion, a moment of inertia $\mathfrak{I}$ governs the absolute energy spacing of the states in the $I(I + 1)$ sequence of the band [108, 109]. This moment of inertia involves both protons and neutrons, and it is a mass moment of inertia. For even–even nuclei the empirical moment of inertia can be related simply to the excitation energy E_{2+} of the first excited state by:

$$E_{2+} = 3\hbar^2/\mathfrak{I} \quad (68)$$

and if the nucleus behaves as a rigid body with axial and parity symmetry, the moment of inertia is directly related to the quadrupole deformation parameter [108] by:

$$\mathfrak{I}_{rig} = \frac{2}{5}MAR^2(1 + 0.31\beta + 0.44\beta^2 + \dots) \quad (69)$$

This value is expected in the case the nucleonic motions in the intrinsic reference frame are those of non-interacting single particles in the intrinsic average potential.

A famous historical figure, Fig. 2 of Ref. [108] reproduced here in Fig. 10, showing the empirical moments of inertia (extracted from known energy band sequences in presumed deformed nuclei with equations of the type of Eq. 68) in units of $\mathfrak{I}_{rig}$ as a function of the deformation parameter β revealed that the flow of nuclear matter during the rotation is neither that of a irrotational fluid, nor that of a rigid rotor. This was interpreted as due to residual interactions between the nucle-

ons considered as single particles in the deformed average potential.

Initially, this was considered an additional victory for the collective model, because for the nucleus to “rotate,” must it not necessarily be deformed? Moreover, this explained the discrepancy between the deformation parameters β obtained from measurements of electric quadrupole moments and those derived from the energy gaps in rotational spectra. However, upon closer inspection, when considering the sequence of states in the rotational band towards higher angular momenta, the moment of inertia is generally not constant. It appears as though the nucleus elongates as the angular momentum increases. This phenomenon, called “nuclear stretching,” was early on associated with a centrifugal effect [110]. Mackintosh considered it like throwing a wrench in the beautiful machinery of nuclear shape [8]: “The usefulness of the concept of static shapes is undermined by the tendency for nuclei to stretch as they rotate. Nearly all methods of measuring nuclear shape, including all the most accurate, involve transitions between rotational states; what then is measured if the states in question do not have the same shape (i.e. as a theoretical entity. The problem is that the theoretical shape is not an accurately measurable empirical quantity)?”.

Nevertheless the situation is clear at that moment of the history: the moment of inertia is, in a way, considered as the embodiment of the usefulness of the concept of nuclear shape, and it has garnered enormous attention. The study of the moments of inertia of deformed nuclei subsequently reached a very high degree of refinement and became a significant source of inspiration for the development of nuclear structure models (see Ref. [111]) and for spectroscopy. Furthermore, the moment of inertia relates energy and total angular momentum, making it the key physical quantity to study to understand at what excitation energy the nucleus adopts a given total angular momentum value (the lowest energy states for a given angular momentum are even given a specially coined term: Yrast¹³). This encompasses the entire question of spin generation in an many-body system like the nucleus, which covers an immense phenomenology gradually uncovered by the commissioning of heavy ion accelerators and the exploitation of fusion-evaporation reactions [93]. Additionally, the revolution brought about by the availability of

¹³ From footnote 1 in Ref. [112]: “By ‘yrast’ level of a given nucleus, at a given angular momentum, is meant the level with least energy at that angular momentum. The English language seems not to have a graceful superlative form for adjectives expressing rotation. Professor F. Ruplin (of the Germanic Languages Department of the State University of New York, Stony Brook) suggested the use of the Swedish adjective yr for designating these special levels. This word derives from the same Old Norse verb heir/a (to whirl) as the English verb whirl, and forms the natural superlative, yrast. It can thus be understood to mean ‘whirlingest’, although literally translated from Swedish it means ‘dizziest’ or ‘most bewildered’”.

high-resolution gamma spectrometers, thanks to advances in semiconductors, played a crucial role [113].

Thus, the strength of the intrinsic shape concept transferred to the idea of an intrinsic structure being set into rotation. This intrinsic structure is of a “microscopic” nature (thus restoring the reductionist honor mentioned in the introduction), the χ_Ω 's in Eq. 64 describe a “shell model” structure or more broadly they correspond to a mean-field description in terms of individual particles that evolves within the boundaries of a nuclear surface. This surface, in turn, is itself subject to motion, primarily of two types—by analogy with molecular physics: rotation and vibration. The interplay between these two types of dynamics, collective and individual (or macroscopic and microscopic, in a sense), is a crucial question for understanding the final chapter of the historical period under consideration, which ultimately led to the discovery of superdeformation—an “exotic” nuclear shape—at high angular momentum (often referred to as “high spin”). It is this last point that we shall now address.

Earlier, we discussed the role of the discovery of fission and its interpretation, highlighting in particular (see Fig. 7) that the most extreme nuclear shapes naturally emerge along the path leading to the division of the fissioning nucleus into two fragments. However, these shapes exist only (in a pure liquid-drop Bohr-Wheeler vision) ephemerally within an essentially dynamic process. By that time, it was already clear that the only mechanism capable of stabilizing such shapes had to go beyond the liquid-drop model—something that would “force the nuclear surface” to distort. As we have just seen, this had to be related to the motion of individual-particle states—and what could be more stabilizing than a shell effect?

Although M. G. Mayer had already suggested in 1948 that the magic numbers 50 and 82 might influence the asymmetry observed in the mass distribution of fission fragments [81], Hill and Wheeler, somewhat surprisingly, dismissed this argument in their legendary paper “*Nuclear Constitution and the Interpretation of Fission Phenomena*” [90] (which we have already discussed extensively here). They did so in a section explicitly titled “*Shell Effects Not an Explanation*” (*sic*). Nevertheless, these same authors introduced into the formal language of nuclear physics the idea that the potential nuclear energy could be a continuous function of deformation (in terms of one or more deformation parameters) and that it could exhibit a primary well, describing the dynamics of the nuclear ground state, but also one (or several) secondary minima, separated from the primary well by a barrier. Transitions between states in this secondary well and those associated with the ground-state structure would thus be hindered, leading to the emergence of *shape isomers*.

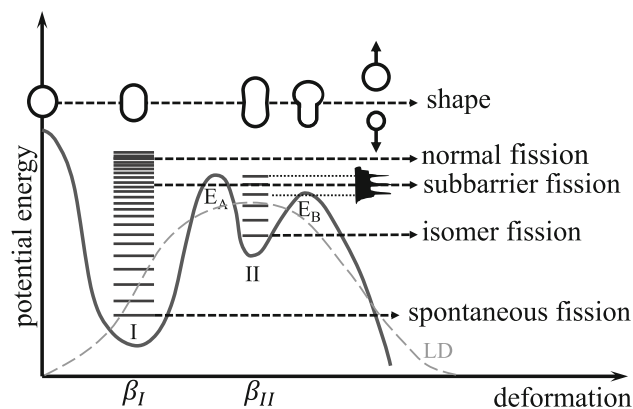


Fig. 11 Figure inspired by Fig. 31 in Ref. [114] displaying a “double-hump” fission barrier, the second potential minimum being due to some specific (deformed) shell effects (see text). The level structure in the main ground-state well is denoted I, and the one in the second well II. The lowest state of this class II structure is a shape isomer. This shape, if axial, has a considerably larger β deformation parameter than the ground-state shape, $\beta_{II} \gtrsim 2 \times \beta_I$ viz. $\beta_{II} \approx 0.65$ (these shapes are shown at the top of the figure). The figure also illustrates two important phenomena which can be explained by the existence of a shape isomer: isomer fission and subthreshold fission (due to the existence of neutron resonances, symbolically represented as horizontal ‘rake teeth’). The gray curve represents (schematically) what the shape of a pure liquid-drop (LD) model fission barrier would look like

This concept remained purely speculative¹⁴ until a series of intriguing discoveries related to fission began to accumulate. The first was the 1961 discovery at JINR, Dubna by Polikanov et al. [116] of an isotope (identified later on as an isomer of ^{242}Am) with an unusually short spontaneous fission half-life of 14 ms.¹⁵ This half-life was several orders of magnitude shorter (a factor 10^{-20} !) than any previously observed. The authors of the discovery suggested that a possible explanation was that this nucleus was actually in an excited state, at an energy level where the fission barrier thickness to be overcome was significantly lower. Strutinsky’s work [118,119], demonstrating that deformed shell effects could lead to the emergence of a second potential well in this barrier, was decisive in understanding the phenomenon. As shown in Fig. 11, the existence of this second well also provided an explanation for other fission-related phenomena that were beginning to accumulate, notably the discovery of subthreshold fission resonances following slow-neutron capture [120,121].

The study of the excitation function of these isomers [122,123] revealed that they were located around 2–3 MeV above the ground state, and further experiments established that they had zero or low spin [124]. The combination of these

¹⁴ According to Ragnarsson et al. [114] the first suggestion that some shell effects could stabilize non-spherical shapes was due to Geilikman [115].

¹⁵ This finding marked the beginning of a long series of spontaneous fission isomer identifications (see e.g. Ref. [117] and Refs. therein).

two observations suggests that this form of isomerism does not correspond to the usual “spin trap” mechanisms—where isomer formation is typically driven by large spin differences and close energy spacing between levels—but rather represents an entirely different type of isomer [125]. The discovery of a rotational band built upon the fission-isomeric state in ^{240}Pu , exhibiting a moment of inertia more than twice that of the ground-state rotational band, provided the first experimental evidence that fission isomers correspond to highly deformed nuclear states [126]. Habs et al. were the first to measure the lifetimes of states in a second potential well (that of ^{239}Pu) and deduced a quadrupole moment value between 34 and 39 eb, the highest quadrupole moment measured at that time. The quadrupole moments of the ground-state bands of certain actinides were already known, but in those cases, the measured values were typically of a dozen of eb. In the abstract of their paper, these authors explicitly stated “The deformation of the second minimum described by the axis ratio of a spheroid is deduced to be $c/a = 2.0 \pm 0.1$ ”. This is schematically illustrated in Fig. 11, the nuclear shape adopted in type-II states is markedly different from that of type-I states. In modern terminology, this phenomenon is now referred to as superdeformation at zero-spin.

In parallel, as previously discussed, the questions surrounding the mechanisms governing the generation of nuclear angular momentum and the origin of the moments of inertia inferred from observed rotational bands were already of significant interest at the time, prompting extensive research and a wealth of literature. In particular, it was expected that for angular momenta beyond $I \approx 20\text{--}30\hbar$ (where nuclear superfluidity effects would be suppressed), the nuclear moment of inertia would approach that of a rigid body with a given shape. Cohen et al. [127] were the first to investigate the evolution of nuclear shape as a function of spin within the framework of the liquid-drop model, considering only the interplay between Coulomb repulsion and surface tension (i.e. without shell effect). At zero spin, the nucleus is naturally expected to be spherical, gradually adopting an increasingly oblate shape as angular momentum increases. Beyond $I \sim 70\hbar$ in the rare-earth region, the nucleus undergoes a sudden shape transition to a triaxial configuration before continuing to elongate into progressively more prolate shapes. Eventually, the fission barrier vanishes, which occurs around $I \sim 100\hbar$ in the rare-earth region and at lower spins for other mass regions.

It was in the mid-1970s that it became clear that the observed axis ratio for shape isomers in the actinides, $c:a \sim 2$, had to be associated with a specific shell effect, which was also expected to manifest in other mass regions [128, 129], particularly in lighter nuclei. In this latter case, the shape was expected to be stabilized by rotation (i.e., at high angular momentum) rather than solely by the Coulomb interaction—a phenomenon more prevalent in nuclei with very high Z ,

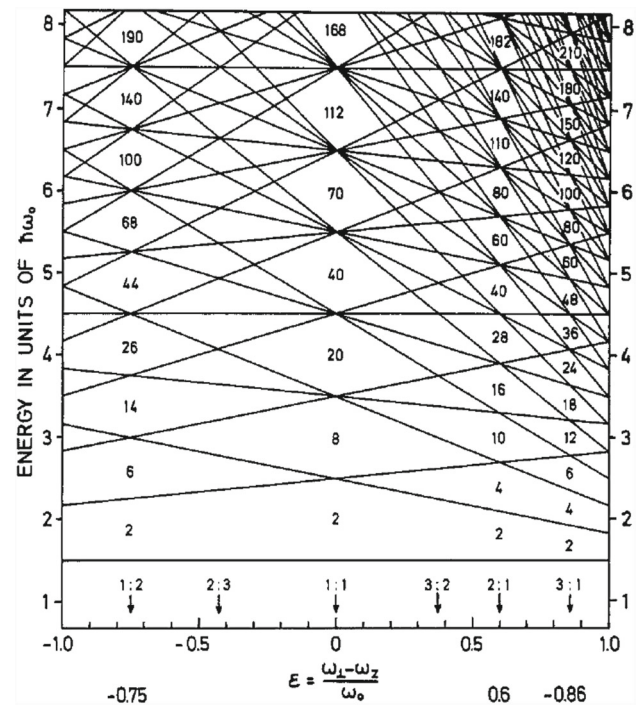


Fig. 12 Single-particle level spectrum of the axially symmetric harmonic oscillator. The oscillator frequencies parallel and perpendicular to the symmetry axis are denoted by ω_z and $\omega_\perp$ respectively, while $\omega_o = \frac{1}{3}(2\omega_\perp + \omega_z)$. The levels are $2(n_\perp + 1)$ degenerate, counting spin degeneracy, where $n_\perp$ is the number of quanta in the perpendicular. Reprinted from Physics Reports, Vol 45, Issue 1, Pages 1–87, 1978, Ingemar Ragnarsson, Sven Gösta Nilsson and Raymond K. Sheline, Shell structure in nuclei, Copyright 2025, with permission from Elsevier (License Number 5959280695062) [114]

such as the actinides and beyond. Thus, at very high spin, the large moment of inertia associated with significant nuclear elongation should render these highly deformed states energetically favored over spherical or moderately deformed configurations. The nucleus ^{152}Dy ($Z = 66$, $N = 88$) was predicted to be a promising candidate for the observation of such high-spin deformation [130].

The formation of shell effects at large deformations, as well as the existence of specific axis ratios that favor these effects, can be illustrated as a first approach using the single-particle state spectrum of a spheroidal harmonic oscillator, as shown in Fig. 12. The spheroidal harmonic oscillator has the advantage that it exhibits remarkably simple analytical properties. For any spheroidal shape with a frequency ratio $\omega_\perp:\omega_z = k/l$, corresponding to an axis ratio $a_\perp:a_z = l/k$, the single-particle levels display degeneracies whenever k and l are integers. This is illustrated in Fig. 12. The propensity to form shell structures in deformed nuclei persists even in more sophisticated nuclear models. Notably, shell effects are most pronounced for small values of k and l , diminishing as these parameters increase. The strongest shell structures emerge for $k:l = 1:1$, corresponding to a spherical shape.

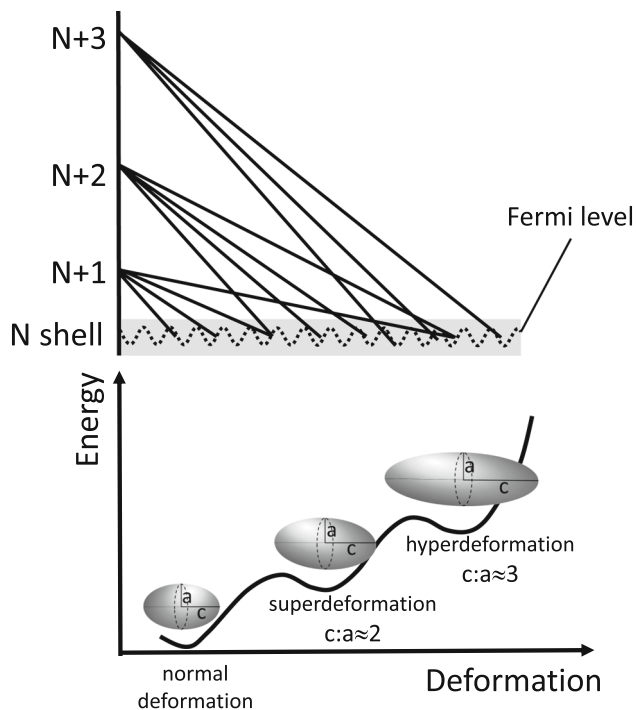


Fig. 13 Pedagogical illustration of the shell-effect driven deformation mechanism. Nuclei with normal deformation correspond to the occupation of “normal intruder” states from a single major oscillator shell above the spherical Fermi surface ($N + 1$), superdeformed nuclei to the occupation of “super-intruders” states stemming from two shells above ($N + 2$), and hyperdeformed nuclei are expected to involve “hyper intruder” states originating from three shells above ($N + 3$)

The deformed configurations with the highest degeneracies occur for $k:l = 1:2$ (oblate shape) and $2:1$ (prolate shape).

One can always envision that if the motion of the nuclear surface is assumed to be purely rotational, the concept of an underlying internal structure in the form of single-particle states remains valid even at high rotational frequencies. The key difference is that, instead of a static single-particle Hamiltonian, one that explicitly accounts for rotational effects must be employed. As a result, rotation directly influences single-particle energy levels, and shell energies can be significantly altered by variations in rotational frequency. In fact, rotational frequency introduces an entirely new degree of freedom, as fundamental as nuclear deformation. Rotation gradually but profoundly reshapes nuclear structure, leading to the breakdown of traditional magic numbers and the emergence of new ones. These ideas were first clearly articulated in the works of Bohr and Mottelson [41, 131]. The first quantitative calculations exploring the full impact of rotation on nuclear structure were conducted by different theory groups at Lund and Warsaw [128, 132], Dubna and Rossendorf [129, 133] and Jülich [134].

Figure 12 clearly illustrates that, in the simple limiting case of the deformed harmonic oscillator, the axis ratio $c:a \sim 2$, favored by deformed shell effects, should manifest

itself regardless of the considered mass region. Spheroidal shapes characterized by this ratio are referred to as “superdeformed”. As previously explained, higher axis ratios can also be favored, but the associated shell effects are expected to become increasingly weaker. Indeed, a clear sequence of energy gaps corresponding to an axis ratio of $c:a \approx 3$ can be distinguished in Fig. 12. However, it is also evident that the amplitude of these gaps is smaller than for $c:a \sim 2$ and even more so compared to the spherical case. A nucleus with a spheroidal shape characterized by $c:a \sim 3$ is referred to as “hyperdeformed”¹⁶. However, realistic nuclear potentials modify this simple picture, and superdeformed nuclei are observed with axis ratios in the range $c:a \sim 1.6$ to 1.8 [135]. Similarly, hyperdeformed nuclei are expected to exhibit $c:a$ values smaller than 3, making a classification based solely on the precise deformation parameter less meaningful.

It turns out that each type of deformation (normal, superdeformed, or hyperdeformed) can be systematically linked to the occupation of specific high- j intruder states [136]. Nuclei with normal deformation correspond to the occupation of intruder states from a single major oscillator shell above the spherical Fermi surface ($N + 1$, “normal intruders”). Likewise, superdeformed nuclei are associated with intruder states from two shells above ($N + 2$, “super-intruders”), and by analogy, hyperdeformed nuclei are expected to involve intruder states originating from three shells above ($N + 3$, “hyper-intruders”) (See Fig. 13).

Following the observation of fission isomers there was a gap of about 25 years during which experimental (allowed by the continuous development of multi detector high-resolution gamma-ray detector arrays) and theoretical results accumulated [93, 113, 137], culminating in the discovery of super deformation in ^{152}Dy [138]: the observation of a high-spin band which can be interpreted as the rotation a spheroidal shape with integer values for the major-to-minor axis ratios (2:1). This breakthrough opened the way for research into even more extreme shapes (hyper deformation), which have yet to be fully realized [139–141]. It is noteworthy that since the historical discovery of irregularities in the isotopic shifts of Sm, the rare earth region has once again proven to be highly conducive to a new discovery related to the concept of nuclear shape.

¹⁶ An amusing distinction, given that the words *super* and *hyper* have exactly the same meaning and strength. They are synonyms: *super* comes from Latin and *hyper* from Greek, both signifying “above”.

3.3 3rd period 1986–now: Shape coexistences and ubiquity

The discovery of superdeformation led to intense experimental and theoretical activity until the late 1990s in the field of high-spin states of nuclei.¹⁷ This activity was supported by advances in on-line gamma spectroscopy devices [142] and theoretical progress, notably through self-consistent mean-field calculations [143, 144]. These research efforts dominated the field and somewhat overshadowed pioneering developments related to the invention of the Isotopic Separation On Line (ISOL) technique for the production of radioactive beams [145, 146] already achieved during the previous period (particularly at ISOLDE/CERN starting in the late-1960s). In this context, we can mention the important example of the pioneering measurements of hyperfine spectra of neutron-rich sodium isotopes using early lasers conducted at CERN by teams from Orsay [147]. These measurements, extended up to mass 31 (^{31}Na has $N = 20$, a magic number of neutrons), combined with direct mass measurements [148], revealed an anomaly compared to expectations as the magic number was approached. Since Townes (as seen in Sect 3.1.2, Fig. 8), it was believed that quadrupole moments cancel out at magic numbers, and a magic nucleus could only be spherical. However, in this isotopic chain, significant increases in isotope shifts and consequently in $\delta\langle r_c^2 \rangle$ were observed as $N = 20$ was approached, accompanied by anomalies in mass defects. Shell model theorists began discussing the “collapse of the conventional shell-model ordering” [149], and one of the most well-established properties of nuclear magicity, sphericity (lack of deformation), seemed to lose its universal character. This is an important event from the perspective of the history of the concept of nuclear shape. The earlier parts of this article have shown that the development of deformation as one moves away from magic numbers played a key role in Rainwater’s invention of the concept of nuclear shape. Here we had a first example showing the deformation could develop while *approaching* a magic number!

Efforts to produce radioactive beams, either at low energy (a few tens of keV, the extraction energy at the ion source) or at high energy after post-acceleration (a few tens of A.MeV, the first post-accelerated radioactive beam, ^7Be , was obtained at Louvain-la-Neuve [150]), would fully bear fruit

¹⁷ Over this period, which was crucial for the development of our understanding of nuclear structure, the author will not elaborate further beyond what has already been discussed—specifically, the contribution of high-spin physics to the development of the nuclear shape concept, which is the main objective of this article. Nevertheless, it is worth noting that during the 1990s, this field reached an extraordinary level of sophistication, accompanied by fascinating discoveries. Readers interested in these developments will find great value in the excellent (and indeed both prolific and proliferating) reviews on the subject (e.g. Ref. [113]) and the extensive literature quoted therein.

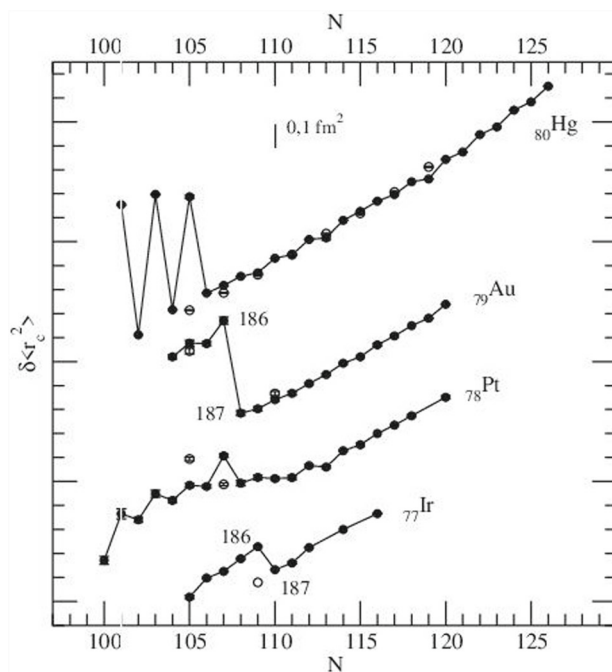


Fig. 14 Mean square charge radius variation obtained from isotope shift measurements in the “transitional region” (isomeric states are represented by open circles), taken from [152]

during this third period. These efforts allowed the extension of measurements that had significantly contributed to the genesis of the concept of nuclear shape—such as hyperfine spectra and Coulomb excitation—to nuclei far from the valley of stability. These topics have been extensively and excellently reviewed in recent literature, so it is unnecessary to delve into the details here. Interested readers can refer to reviews by Zielińska [52], Cheal and Flanagan [151], Campbell, Moore and Pearson and [38], and by X.F. Yang et al. [17], and the abundant literature cited therein.

The beginning of the 1970s was marked by another absolutely exceptional discovery, made in a region of the nuclear chart, nestled between the “good rotors” of the rare earth elements and the transuranic elements: the so-called “transition region”, a region which was likely somewhat problematic during the previous period.¹⁸ Ernst Otten¹⁹’s group in Mainz, exploring the hyperfine structure of neutron-deficient mercury isotopes, discovered a gigantic oscillation in the isotopic shifts between isotopes with even mass numbers and those

¹⁸ Mariscotti et al. noted that “The stable Pt nuclei with $A = 190, 192$, and 194 still possess appreciable moments of inertia and large but ‘finite’ softness parameters. Hence they may be characterized as ‘pseudospherical’”, as they developed their variable moment of inertia model. Incidentally, the study of Ir nuclei belonging to this region, known to be complicated from the perspective of the nuclear shape concept, was indeed the subject of the author’s PhD thesis [152, 153].

¹⁹ The author had the distinct honor of having Ernst Otten as a member of his thesis committee.

with odd mass numbers, starting from $A < 186$ [154]. This change, interpreted as the transition from a slightly oblate shape along the symmetry axis to a well-elongated shape between $N = 106$ and $N = 105$, occurs then again successively between $N = 104$ and $N = 103$, and between $N = 102$ and $N = 101$, thus giving rise to an anomalous “odd–even staggering” of the mean square charge radius [155] as shown in Fig. 14. Generally, a small amplitude oscillation of the mean square charge radius between odd and even isotopes is observed throughout the table of nuclides [156]. In the so-called normal case, it corresponds to a larger mean square charge radius $\langle r_c^2 \rangle$ for the even nucleus compared to the odd nucleus which is usually not accompanied by a change in the nuclear shape. The variation of the mean square charge radius $\delta \langle r_c^2 \rangle_{\text{even-odd}}$ measured here were both the largest ever observed at that time [155] and inverted compared to the general case: the odd nucleus is larger than the even nucleus.²⁰ Another striking feature of the mercury isotopic series is the very large difference in the mean square charge radius between the ground state $I = \frac{1}{2}^-$ and the isomeric state $I = \frac{13}{2}^+$ of the ^{185}Hg nucleus, which is exceeded only in the case of the previously discussed fission isomers (see Sect. 3.2). This was also interpreted as due to the fact that the ground and isomeric states, though very close in energy, were possessing very different shapes [155]. The rest of this region, extending towards lower Z values, was systematically explored using laser spectroscopy, primarily at ISOLDE/CERN, until $Z = 77$ (Ir) was finally reached at the end of the 1990s, see [152, 153] and references therein. The extraordinary discoveries made in the measurements of charge radii in neutron-deficient mercury isotopes naturally motivated numerous other studies in this region, particularly spectroscopic ones [162] (which would culminate in the identification of a triple shape coexistence in ^{186}Pb [163]). In line with the prevailing approach of previous periods, it was considered necessary to cross-check observables from different sources (hyperfine spectroscopy, gamma spectroscopy, conversion electrons, Coulomb excitation, etc., see Fig. 15) to

²⁰ To better illustrate the concept of odd-even staggering, a phenomenon that is almost universally present in isotope shift measurements across all accessible isotopic chains, one can quote Angeli et al. [157]: “The odd-even staggering is a phenomenon observed in practically all optical isotope shift experiments. However, this effect is in most cases disturbed by shell closure or deformation, resulting in values differing strongly not only in magnitude but also in sign: anomalous odd-even staggering. Indeed, the different effects are not simply superimposed, but in some cases a strong interference between odd-even staggering and deformation is observed, e.g. [158–160]: the pairing interaction ‘triggers’ the shape change. Experimental studies of the isotopic radii trend of different elements usually discuss this effect and there is a large amount of papers devoted to the odd-even staggering of separated elements often in restricted isotopic chains. As regards the theory, it is shown [161] that with a suitable three-body part in the effective interaction, one is able to describe the odd–even staggering of charge radii along isotopic series of Sn and Ba.”

attempt to uncover an underlying reality conceptualized in terms of nuclear shapes (Ref. [153] is likely one of the last papers truly written in this spirit). Interpretations spoke of “stabilization of the overall nuclear shape conferred by the unpaired nucleon,” exactly in the spirit of Rainwater.

Concepts of “shape transition” and “shape coexistence” began to crystallize²¹ (for example, the transition from one shape to another between a mercury isotope and its neighbor, which we just mentioned, falls into the first category, while the significant shape difference between the ground and isomeric states of ^{185}Hg belongs to the second). Actually these concepts had been “in the air” for a very long time: the phenomenon of shape isomers, which we discussed earlier, can to some extent be considered as falling into the category of shape coexistence. W. Nazarewicz considered [165] that Fig. 37 of the Hill and Wheeler 1953 paper [90] already announced the very idea of shape coexistence but also pointed out that they did not consider this example to be realistic: “the figure shows an idealized case — not expected to occur in actual nuclei” [90]. As early as in 1956, H. Morinaga [166] interpreted the $I^\pi = 0^+$ state located at 6.049 MeV excitation energy in ^{16}O as the head of a 0^+ , 2^+ , 4^+ , ... rotational band with a 4-particle-4-hole ($4p-4h$) structure (a 4 hole state in ^{20}Ne). Another case considered as an early example of multiple-shape coexistence [165] is ^{40}Ca which exhibit two clearly rotational bands associated with $4p-4h$ and $8p-8h$ states at 3.352 MeV and 5.213 MeV, respectively [167]. The literature by Kumar [168] already mentioned phase transitions as early as the beginning of the 1970s. However, it was truly during the period we are discussing here that shape transition and coexistence phenomena were identified, recognized, and studied as such—forming a conceptual framework linking a vast network of observables spanning the entire nuclear chart (see e.g. Figure 8 in Ref. [162]). The emergence of the concept of shape coexistence as we know it today would not have been possible without the foresight of K. Heyde and J.L. Wood, who, sometimes with other authors, dedicated several reviews to this topic [55, 162, 169, 170]. Their latest one [55], often taking on a testamentary tone, particularly illuminates the genesis of the concept of shape coexistence and outlines the major phenomenological lines that led to it. Additionally, the more specific reviews by G6rgen and Korten [171], as well as Gade and Liddick [172], are recommended for their insights into recent experimental

²¹ For instance, A. V. Ramayya, E. F. Zganjar and W. T. Pinkston wrote in 1992 in their dedication to J.H. Hamilton at the Sanibel Symposium [164]: “Joe Hamilton has played a key role in exploring the physics of nuclear shapes and nuclear shape coexistence. In 1986, at an international conference in Oak Ridge, Professor Amand Faessler credited Joe with changing our view of deformed nuclei. Instead of having one fixed shape (spherical or deformed), in reality nuclei are more complex; energy bands associated with different shapes coexist within the same nucleus in the same energy range.”

developments in this field. It should be noted indeed that the emergence of this new approach to exploring the concept of nuclear shape was made possible by the coherent interpretation of observables obtained through entirely and profoundly independent methods: masses, charge radii, excitation energy spectra, nuclear state lifetimes, electromagnetic transition probabilities, etc. All of these data require the implementation of very different and diverse experimental techniques.

Heyde et al. [173] took care to analyze (and thus define) the difference between phase transitions and shape coexistence. According to them, the distinction is related to the fact that the number of basis states is preserved through a phase transition along an isotopic or isotonic chain as the involved structure progresses from one limiting case into the other: “the eigenstates of one limit will be spread out and finally end up as the eigenstates of the other limit” [173]. They also specify that [173]: “on the other hand, phase coexistence can appear when a number of basis states, appearing at very high excitation energy (outside of the model space that is regularly considered as the space of low-lying configurations) under normal circumstances (such as particle-hole excitations across closed shells), can profit from residual proton–neutron interactions. This can be the case in nuclei with $a(n)$ (almost) closed shell of protons (or neutrons), on one hand, and a large number of valence neutrons (or protons), on the other hand. In such a situation, the proton–neutron correlation energy in the p-h excited high-lying configurations can become competitive with the energy needed to create this family of extra states and so see the energy drops towards the low-energy regime. Such classes of states are also quite often called ‘intruder’ excitations”. The observations made in the Hg isotopic chain, discussed earlier, fall therefore under that last category of shape coexistence according to their criteria: two shapes coexist in neutron-deficient Hg nuclei (p-h states stem out from proton jumps across the $Z = 82$ shell gap), alternating their presence in the ground state depending on whether the nucleus has an even or odd number of neutrons.

From the perspective of the history of the concept of nuclear shape, this period is therefore marked by a complete dissociation of the applicability of the shape concept from earlier hesitations. Unlike the previous period, where the concept of nuclear shape was almost inherently tied to the collective model and considered applicable only to what we would nowadays call “good rotors,” the manifestation of nuclear shapes is now considered virtually universal. And it is not anymore restricted to a spin-generation related question, because thanks to the advent of laser collinear spectroscopy techniques many unexpected shape-related features appear to occur in (β -unstable) ground states, or long lived isomers. Nuclear deformation appears in unexpected places—that would surprise its original proponents indeed!—like at magic numbers associated with spherical shell closures. Ultimately, the emerging image at the end of this period, follow-

ing Heyde and Wood [55], is that nuclear deformation is always present. All nuclei exhibit multiple states of deformation, multiple shapes; it just so happens that the interplay of correlations and the inhomogeneities in the energy distribution of individual particle states may favor one shape over another. Typically, at shell closures, the spherical shape is preferred (when it is not we have an “island of inversion”), and far from proton and neutron shell closures, a quadrupole axial shape can also be favored, especially for very large shells. Most importantly, at a time when *ab initio* calculations are making immense progress, the nuclear shape is seen as just another emergent phenomenon in a strongly correlated many-body system [3]. This represents a complete paradigm shift from the previous period when the concept of shape was seen as operational and useful.

4 Conclusions and perspectives in light of recent results from ultra-relativistic heavy ion collisions

The shape of the atomic nucleus eludes us and will continue to do so simply because it necessitates the introduction of an intrinsic, body-fixed reference frame attached to the nucleus, and one cannot transport oneself onto the nuclear surface. There is no operational definition of an observable that can measure the deformation or shape of the nucleus. Thus, the nuclear shape is primarily a concept-one derived from our everyday experience and applied to nuclear physics, introduced by the great pioneers of our field as an elegant way to reconcile nuclear properties that were deemed incompatible or even contradictory.

As we have seen throughout this article, the introduction of this concept was not without formal difficulties (such as the issue of redundant coordinates) and paradoxes. This concept hails from a time when “vivid pictures” were unhesitatingly employed to solve the formidable problems posed by the atomic nucleus, a time when nuclear spectroscopy was enchanted with shapes and shells.

Today, the approach is more rigorously reductionist, and experimental results (which must imperatively be reduced to calculable observables) are merely compared with computational results rooted in increasingly fundamental principles, ideally devoid of any reliance on these old, outdated images (the author himself used to introduce an empirical moment of inertia in his microscopic calculations in the early 2000s [153]; such methods are, of course, now proscribed). Yet, we continue, perhaps surprisingly, to reference nuclear shapes, and even use them as motivation: the search for “exotic” shapes (hyperdeformation and Jacobi shape transition [174, 175], tetrahedral symmetry [176] or the pear, parity-breaking shape [24] as touched upon in Sect. 2.4, to name a few) remains a vibrant branch of nuclear physics.

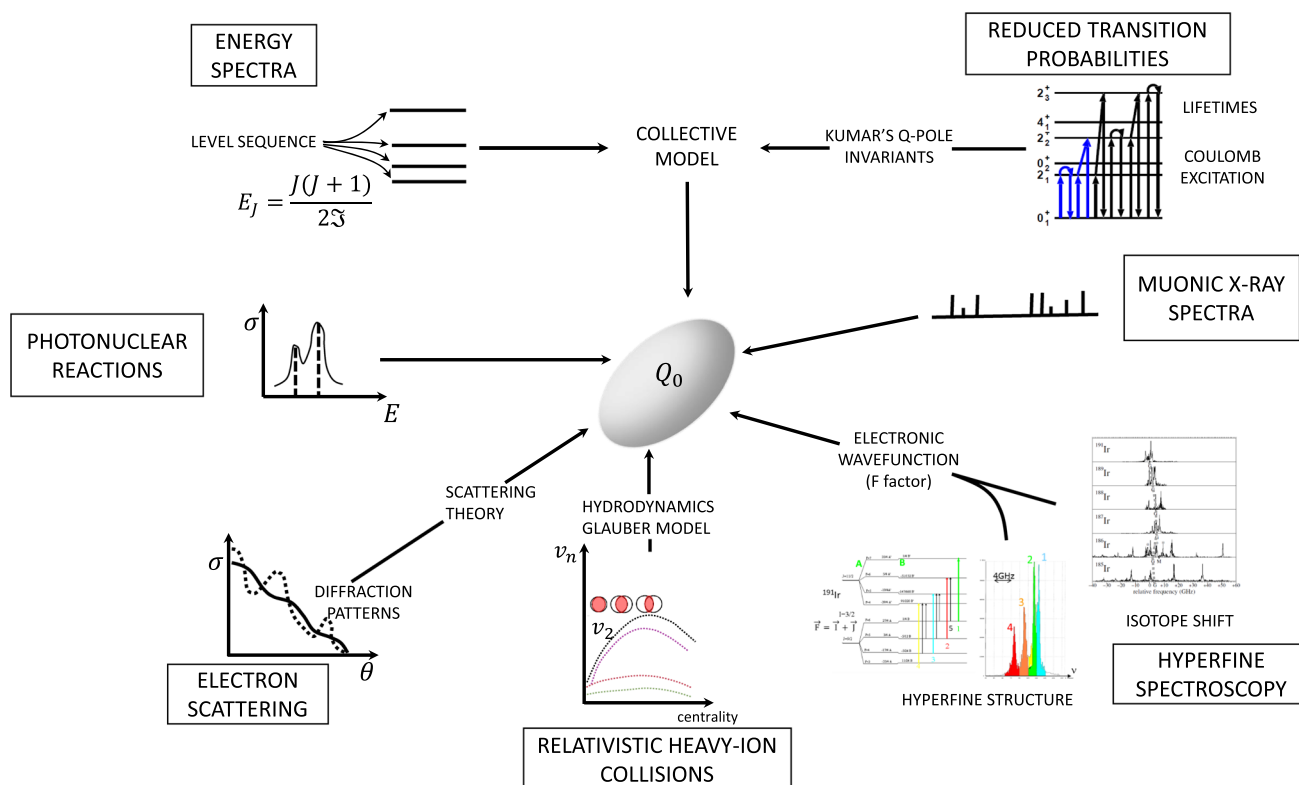


Fig. 15 This figure is inspired by Fig. 1 in Ref. [4], which schematically presents the vast array of experimental data that can be effectively interpreted by attributing an intrinsic shape to nuclei or that

point towards its existence (represented here by the intrinsic quadrupole moment Q_0 of a spheroid) and the models and/or concepts required to establish the connection

It is clear that it is tempting to believe in nuclear shape as an underlying, albeit unattainable, reality because the concept allows for the correlation of an impressive number of observations from entirely different sources and types of measurements. This was demonstrated as early as the first review on the subject by Temmer, with his beautifully crafted Fig. 1, [4], for which a revisited version is offered in Fig. 15.

The major classes of experiments discussed in the present article, focusing on the genesis and evolution of the concept of nuclear shape, have been deliberately limited to those related to hyperfine spectroscopy and the collective model. However, compared to Ref. [4], a new branch has been introduced into this landscape: ultra-relativistic heavy ion collisions, which emerged in the early 2020s [5–7]. Few recent contributions to the concept of nuclear shape have come from the branches of photo-nuclear reactions (splitting of the Giant Dipole Resonance peak), electron scattering experiments (filling the diffraction minima due to nuclear translucency), or studies of muonic atom X-ray spectra (nondiagonal interactions between meson and deformed nucleus leading to the possibility of excitation of some of the low-lying nuclear rotational states during the cascade). It should be emphasized that these experiments reached an extraordinary

level of refinement and clearly helped establish and develop the concept of nuclear shape throughout history. Unfortunately, they fell into disuse mainly during the 1990s. Readers interested in the essential results from these three branches can profitably refer to Mackintosh’s review [8].

But let us return to the significant novelty of Fig. 15: relativistic heavy ion collisions. It turns out that to achieve the correct spatial correlations in the initial moments of quark-gluon plasma formation, it is necessary to conceive of a collision between two nuclei possessing an intrinsic shape. This is particularly evident in the case of ^{238}U on ^{238}U collisions [7], where a value $\beta \approx 0.3$ must be assigned—a value very consistent with everything known about these nuclei from other (historical) methods depicted in Fig. 15. This result is quite remarkable as it tends to demonstrate the incredible consistency of the nuclear shape concept, which remains valid across energy scales, including in an energy regime where only the internal degrees of freedom within nucleons would seemingly be relevant.

It is well known in nuclear physics, and has been for a long time, that heavy-ion reaction cross-sections exhibit sensitivity to the deformation of the collision partners. This is particularly true for reactions that are highly sensitive to the

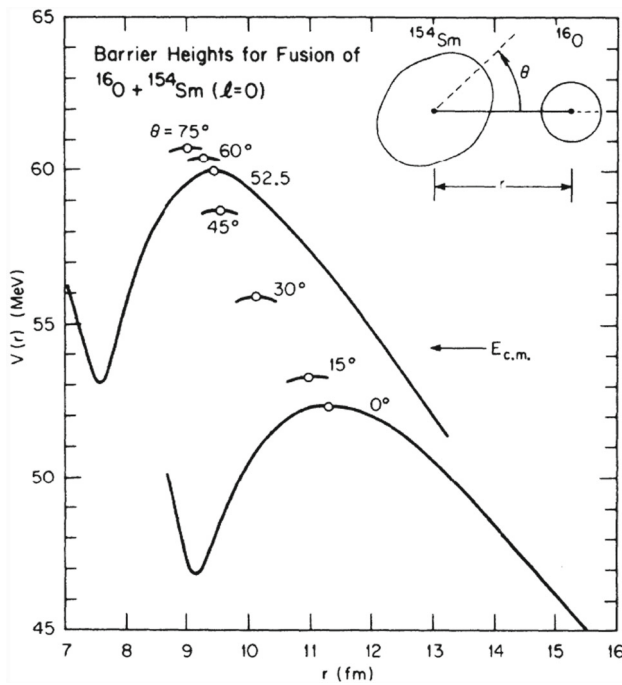


Fig. 16 Calculated fusion barriers and barrier heights in a collision between a low-energy (≈ 60 MeV) spherical ^{16}O projectile and a well axially deformed ^{154}Sm target, as a function of θ , the relative angle between the symmetry axis of the target nucleus and the projectile bombarding axis [177]. Reprinted figure with permission from R. G. Stokstad and E. E. Gross, Phys. Rev. C 23, p. 281 (1981) Copyright 2025 by the American Physical Society (License Number: RNP/25/FEB/088271)

details of the nuclear surface structure, as is the case for fusion reactions around the barrier [178, 179].²² Figure 16 shows very nicely how one may expect the fusion barriers and barrier heights to evolve as a function of θ , the relative angle between the symmetry axis of the target nucleus and the projectile bombarding axis in a collision between a low-energy (≈ 60 MeV) spherical ^{16}O projectile and a well axially deformed ^{154}Sm target [177]. In a classical description one has a barrier height $B(\theta)$ [178]:

$$B(\theta) \approx \frac{Z_1 Z_2 e^2}{4\pi\epsilon_0} \frac{1}{R_B + R_T \beta Y_{20}(\theta, 0)}, \quad (71)$$

where R_B is the average barrier radius, R_T is the average target radius and β is the axial quadrupole deformation parameter as defined previously.

²² One of the authors of Ref. [178], N. Rowley, even suggested to use the experimental barrier distributions defined as:

$$D_{\text{exp}}(E) = \frac{d^2(E\sigma_{\text{fusion}})}{dE^2}, \quad (70)$$

(where E is the incident energy and σ_{fusion} is the measured fusion cross section) to determine the shape parameters of nuclei.

The scattering of spherical projectiles by deformed nuclei was studied a long time ago [180] in an extreme adiabatic approximation that would correspond to the nuclear translucency picture. This model assumes that the rotational motion of the axially symmetric target nucleus is very slow in comparison with the relative motion of the projectile and target nuclei. Thus, one can evaluate an orientation-dependent scattering amplitude for the scattering of the projectile by the target whose symmetry axis is oriented at angles θ with respect to the beam axis. The elastic and inelastic scattering amplitudes can be obtained by folding this orientation-dependent scattering amplitude with the distributions of orientations θ . In this context, the total fusion cross section is given by [181]:

$$\sigma_{\text{fusion}}^{\text{total}} = \int_0^{\pi/2} \sin \theta \sigma_{\text{fusion}}(\theta) d\theta, \quad (72)$$

where σ_{fusion} is the cross section of the fusion of the projectile. If one considers that this axially deformed target is an even-even nucleus, its ground state wavefunction ($I = 0$) possesses components corresponding to all orientations. Conversely, the wavefunction corresponding to a fixed orientation must possess components corresponding to all angular momenta. Therefore, in the language of quantum mechanics, this classical description must be intimately related to the excitations of the rotational states of the deformed target.

At the energy regime where these fusion studies are conducted, this extreme adiabatic approximation is not valid, and it is necessary to explicitly include excited states of the nucleus in the description (the so-called coupled-channel calculations). However, there is no doubt that the rotation of the nucleus is very slow (even totally meaningless) compared to the collision velocities involved in the ultra-relativistic heavy-ion collisions discussed in Refs. [5–7]. Thus, this invites reflection on part of the interpretation made by the discoverers of the influence of the sensitivity of ultra-relativistic collision data to the shape of the collision partners (definitely a significant discovery for the history of the nuclear shape) related to the possibility of distinguishing, in events where the two nuclei collide with maximal spatial overlap, “tip-tip” collisions (where the major axes of the two spheroids align) from “body-body” collisions (where the minor axes of the two spheroids align). If this is correct, it would be entirely unprecedented: all other measurements shown in Fig. 15 involve a total indeterminacy of the orientation of the intrinsic system, manifested by the phenomenon of nuclear translucency explained in Sect. 2. Hofstadter, who also bombarded the nucleus with an ultra-relativistic probe (electrons are very quickly relativistic particles), explained in his first review in 1956 [12]: “In interpreting the experimental curves, it is necessary to include the effect of an ellipsoidal shape and to average it appropriately in all the aspects seen by approaching electrons. The averaging has the effect of rounding-off the nuclear surface and making the apparent

surface thicker than it actually is”. It seems to the author that distinguishing the relative orientations of the intrinsic shapes of the colliding nuclei is not possible outside the principle of nuclear translucency, regardless (or because) of the “suddenness” of the collision. Quantum mechanics dictates that the quadrupole moment (or any higher-order moment) can only be observed through a transition to a $I = 2$ state, whose excitation energy corresponds to the rotation frequency of the intrinsic quadrupole shape reflecting the fact that the intrinsic reference frame of the nucleus is not an inertial frame for the entire universe. The characteristic interaction times in relativistic collisions range from 10^{-24} s at RHIC to 10^{-26} s at the LHC, several orders of magnitude below the characteristic times of individual particle movements within the nucleus or its collective motion, Eqs. 62 and 63. It is clear, then, that collisions occur between nuclei in their ground states; nothing “rotates” the nucleus at the moment of collision, and therefore nothing can specify the orientation of the intrinsic reference frame.

This point, which may well be mysterious only to the present author, in no way diminishes the fact that the advent of a new probe for nuclear shape constitutes a historic event (over six decades have passed between Fig. 1 in Ref. [4] and Fig. 15 in this article!). Finally, let us not forget that, as always with nuclear shape, it is probably all a matter of... interpretation.

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